

Extracellular vesicles purified from serum-converted human platelet lysates offer strong protection after cardiac ischaemia/reperfusion injury

Dora Livkisa^a, Tzu-Hsin Chang^b, Thierry Burnouf^{a,b,c,**}, Andreas Czossek^b,
Nhi Thao Ngoc Le^a, Gleb Shamrin^d, Wei-Ting Yeh^e, Masao Kamimura^f, David J. Lundy^{a,b,g,*}

^a International PhD Program in Biomedical Engineering, College of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan

^b Graduate Institute of Biomedical Materials & Tissue Engineering, College of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan

^c International Program in Cell Therapy and Regenerative Medicine, College of Medicine, Taipei Medical University, Taipei, Taiwan

^d Ph.D. Program for Cancer Molecular Biology and Drug Discovery, College of Medical Science and Technology, Taipei Medical University, Taipei, Taiwan

^e School of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan

^f Department of Medical and Robotic Engineering Design, Faculty of Advanced Engineering, Tokyo University of Science, Japan

^g Center for Cell Therapy, Taipei Medical University Hospital, Taipei, Taiwan

ARTICLE INFO

Keywords:

Platelet
Extracellular vesicle
Nanocarrier
Myocardial infarction
Biodistribution
Bioprocessing

ABSTRACT

Extracellular vesicles (EVs) from cultured cells or bodily fluids have been demonstrated to show therapeutic value following myocardial infarction. However, challenges in donor variation, EV generation and isolation methods, and material availability have hindered their therapeutic use. Here, we show that human clinical-grade platelet concentrates from a blood establishment can be used to rapidly generate high concentrations of high purity EVs from sero-converted platelet lysate (SCPL-EVs) with minimal processing, using size-exclusion chromatography. Processing removed serum carrier proteins, coagulation factors and complement proteins from the original platelet lysate and the resultant SCPL-EVs carried a range of trophic factors and multiple recognised cardioprotective miRNAs. As such, SCPL-EVs protected rodent and human cardiomyocytes from hypoxia/re-oxygenation injury and stimulated angiogenesis of human cardiac microvessel endothelial cells. In a mouse model of myocardial infarction with reperfusion, SCPL-EV delivery using echo-guided intracavitary percutaneous injection produced large improvements in cardiac function, reduced scar formation and promoted angiogenesis. Since platelet-based biomaterials are already widely used clinically, we believe that this therapy could be rapidly suitable for a human clinical trial.

1. Introduction

1.1. Pathogenesis of myocardial infarction and ischaemia/reperfusion injury

During myocardial infarction (MI) the myocardium is deprived of oxygen and respiratory substrates. Cardiomyocytes (CMs) are highly sensitive to oxygen levels and rapidly begin to die by apoptosis, ferroptosis, necrosis and other mechanisms following hypoxia [1]. The current best practice therapy for MI is rapid restoration of blood flow, which reduces the infarct size but causes a reperfusion injury due to sudden influx of oxygen, pH correction and reactive oxygen species generation [2]. Adult mammalian CMs have almost no capacity for

self-renewal, thus CMs lost during ischaemia/reperfusion (I/R) injury cannot be replaced [3]. Instead, dead CMs are replaced by non-contractile fibrotic scar tissue. Cardiac function may be preserved by reducing CM apoptosis, controlling inflammation, modulating fibrotic remodelling, and promoting microvessel angiogenesis [4,5]. To this end, growth factors, cytokines, microRNAs, small molecules and direct transplantation of therapeutic donor cells have all been explored as potential therapies, with mixed results [6–11].

1.2. Extracellular vesicles in treatment of myocardial infarction

Extracellular vesicles (EVs) are phospholipid bilayer-bound nano-scale vesicles which are secreted by cells as a form of cellular paracrine

* Corresponding author. International PhD Program in Biomedical Engineering, College of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan.

** Corresponding author. International PhD Program in Biomedical Engineering, College of Biomedical Engineering, Taipei Medical University, Taipei, Taiwan.

E-mail addresses: thierry@tmu.edu.tw (T. Burnouf), dlundy@tmu.edu.tw (D.J. Lundy).

<https://doi.org/10.1016/j.biomaterials.2024.122502>

Received 27 September 2023; Received in revised form 6 January 2024; Accepted 5 February 2024

Available online 6 February 2024

0142-9612/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

signalling [12]. As such, EVs contain a diverse array of miRNAs, proteins and other bioactive molecules which can be exploited for therapeutic purposes [13]. EVs isolated from bone marrow, adipose or umbilical mesenchymal stromal cells (MSCs), cardiosphere-derived cells (CDCs), progenitor cells, and induced pluripotent cell (iPSC)-derived cells have all shown benefits in models of MI [14–16]. However, the EV cargo, and subsequent function, varies significantly depending on the original donor, culture conditions, EV isolation methods and many other variables [17,18]. There are also significant financial and logistical challenges in scaling up EV production by culturing therapeutic cells under good manufacturing practices, coupled with the need for vast amounts of xeno-free or EV-free media to generate cell-derived EVs [19].

1.3. Human blood as a source of therapeutic EVs

Studies have shown that EVs from both plasma and serum have therapeutic benefits in animal models of MI, improving ejection fraction and reducing scarring [20,21]. However, there are limitations to using autologous EVs derived from patients' blood; EV cargo and function varies between individuals due to physiological conditions, including exercise, diet, age and the presence of disease [22–24]. Platelet-derived EVs (P-EVs) are often considered as unwanted contaminants in plasma EV studies (i.e. to identify biomarkers), and protocols have been designed to reduce platelet activation or remove P-EVs from blood-derived samples [25,26]. However, recent studies posit that around 50% of circulating EVs are originally derived from platelets and megakaryocytes, their precursor cells [27]. Perhaps due to their reputation as unwanted contaminants, the therapeutic potential of P-EVs remains much less explored than EVs derived from cell culture or plasma/serum [28].

1.4. Uses and advantages of platelet-derived EVs

Allogeneic platelets and their derivatives have many advantages as therapeutic biomaterials. Firstly, human platelet-based therapeutics such as platelet concentrates (PCs) and platelet-rich plasma are already widely-used in the clinic [29,30]. P-EVs are naturally released from platelets *in vivo*, and during the 5–7 day shelf-life period of PCs at 20–24 °C [25]. Therefore, we could consider that P-EVs are already administered to patients when they receive PC infusions, albeit in an un-purified form. Since platelets are anuclear, there is no risk of unwanted proliferation or mutagenesis following administration; thus, the overall safety risks of P-EVs are likely to be inherently low. Secondly, methods for platelet collection, purification, viral screening, pathogen reduction treatments, storing and transporting are all well-established and licensed by national regulatory authorities. This means that sterile, clinical-grade PCs can be obtained at high purity from blood establishments [28]. These PCs can be used to prepare platelet lysates and generate large quantities of P-EVs with less complexity than cell culture-derived EVs [31]. PCs are also routinely pooled from multiple donors before administration to patients, therefore individual donor variations are less impactful on the final product.

From the biological viewpoint, platelets are not just involved in blood coagulation; they are early-responders to injury which participate in tissue repair, releasing trophic factors such as stromal cell-derived factor 1 (SDF-1), vascular endothelial growth factor (VEGF), insulin-like growth factor (IGF) platelet-derived growth factor (PDGF) and others [30]. Upon arrival at an injury site they can also induce differentiation of endothelial progenitor cells [32]. Platelet alpha granules and P-EVs contain multiple cytokines, chemokines, anti-oxidants and growth factors [28]. These factors are known to be bioactive and P-EVs have been demonstrated to improve wound healing outcomes. A first-in-human trial has recently been conducted, which showed no safety concerns using allogeneic PC-derived P-EVs [33]. In the cardiac field specifically, platelets have been implicated in the response to MI and have been shown to accumulate in the heart following MI [34,35].

Illustrating this, nanoparticles coated with platelet-derived membranes showed improved cardiac homing, even several days following MI [35, 36].

1.5. Rationale and hypothesis of this study

We hypothesised that P-EVs generated from pooled human PCs could have cardioprotective effects. To explore this and make our study as translationally relevant as possible, we first aimed to develop and validate an optimal protocol for P-EV production and high purity separation from a sero-converted platelet lysate (SCPL) pool made from human clinical-grade PCs. These were obtained from 15 to 18 healthy donors, a number regarded as sufficient to ensure standardisation [37]. Pooled platelet lysates were produced and sero-converted to generate P-EVs and remove bulk coagulation factors like fibrinogen [38]. Size exclusion chromatography was used to isolate EVs, since this has been shown to isolate high purity EVs from plasma-based samples and remove more contaminating proteins than centrifugation or precipitation-based methods [39,40]. The isolated EVs were tested on human and rodent cells using *in vitro* models of CM protection, cardiac angiogenesis and macrophage polarisation, compared against the original sero-converted platelet lysate. Since EV generation and isolation methods influence EV cargo, we carried out detailed P-EV proteomic and miRNA cargo analyses to examine enriched and depleted factors resulting from our specific methodologies. Lastly, we sought to test the efficacy of isolated P-EVs using an *in vivo* model of cardiac I/R injury, and to carry out biocompatibility assessments.

2. Materials & methods

2.1. Generation of sero-converted platelet lysate (SCPL)

Institutional Review Board approval was obtained from Taipei Medical University (TMU-JIRB N201802052) for use of human materials. Clinical-grade human platelet concentrates (PCs), anticoagulated using a citrate-phosphate-dextrose solution, were obtained by whole blood donations from voluntary, non-remunerated, donors by the Taipei Blood Center, Taiwan Blood Service Foundation (Guandu, Taiwan). On the day of expiration for use in transfusion (5 days following collection), PCs were delivered to our laboratory within 90 min, under temperature-controlled conditions, pooled, then aliquoted. For this study, three donor PCs were processed into sero-converted platelet lysate (SCPL) by the following protocol: a ratio of 0.3 ml 0.9 M CaCl₂ with 5 g of sterile glass beads were added to each 10 ml of PC to induce sero-conversion. The mix was rotated at room temperature until visible clot formation, after which the rotation was continued for an additional 30 min. The mixture was then centrifuged at 6000×g for 30 min. The supernatant (SCPL) was collected, aliquoted, then stored at –80 °C until further use. Each “batch” refers to one run of SCPL generation and chromatographic separation.

2.2. Isolation of extracellular vesicles

EVs were isolated from the whole SCPL (SCPL-W) using size exclusion chromatography (Izon, qEV original 35 nm Gen 2). The column was first equilibrated following the manufacturer instructions. Each batch of SCPL was centrifuged at 6,000×g for 10 min to remove any precipitate, then 500 µl of SCPL supernatant was added to the column. 0.1 µm-filtered PBS was used to elute 30 fractions in total, 0.5 ml each. For comparison, a commercial reagent (ExoQuick, System Bio, EXOQ5A-1) was used for EV isolation, following the manufacturer's protocol. For BM-MSC EV isolation, conditioned medium was first centrifuged at 500×g (10 min, 4 °C), then 3,000×g (20 min, 4 °C), then passed through a 0.22 µm filter. EVs were isolated by ultracentrifugation at 100,000×g (16 h, 4 °C). The pellet was thoroughly resuspended in 0.22 µm-filtered PBS and centrifuged again at 100,000×g (70 min, 4 °C).

2.3. Characterisation of size exclusion chromatography fractions

The protein concentration in each fraction was first approximated by spectrophotometry at 280 nm. Following fraction pooling, protein concentration was quantified using a BCA Protein assay kit (Thermo, 23227). Particle count was determined by nanoparticle tracking analysis (NTA) using a NanoSight NS300 (Malvern). For NTA, samples were diluted with 0.22 µm-filtered PBS to a range of 50–200 particles per frame. Each sample was measured three times for 60 s, then analysed with NanoSight Software (NTA 3.4). Dynamic light scattering (DLS) was used to analyse particle size distribution, using a Zetasizer Lab system (Malvern) and Zetasizer Ver.7.13 software. For DNase treatment, SCPL-W was incubated with TURBO DNA-free kit (Thermo, AM1907), 4U/50 µl, for 30 min at 37°C. DNase inactivation reagent was added and incubated for 5 min, followed by centrifuging at 10,000×g for 90 s. Total DNA was extracted using DNeasy Blood & Tissue kit (Qiagen, 69504) then quantified by UV spectroscopy or Quant-iT assay (Thermo, Q33120). For RNase treatment, SCPL-W was incubated with RNase A (Thermo, EN0531) at 40 µg/ml for 30 min, 37°C. RiboLock RNase Inhibitor (Thermo, EO0381) was then added (1 U/µL). Total RNA was extracted using Trizol LS reagent (Invitrogen, 10296028) and chloroform in combination with RNeasy kit (Qiagen, 74104). The aqueous phase was mixed with 2.5 vol 99.5% ethanol and transferred to RNeasy MinElute Spin columns. The manufacturer's instructions were followed for further RNA isolation steps. mRNA was quantified from 10 µl samples using Quant-It RNA Assay kit (Thermo, Q33140) and read using a Thermo Varioskan Flash plate reader at excitation/emission 644/673 nm.

2.4. Validation of EV protein markers and morphology

Known EV surface and cargo proteins were detected by a human-specific antibody membrane array (System Biosciences, EXORAY210B-8). EVs were first concentrated using Amicon® Ultra-4 Centrifugal Filter unit (Merck, UFC8050), centrifuging the sample at 4,000×g for 20 min. 50 µg of sample protein was suspended in the provided lysis buffer, and the manufacturer protocol for labelling, removal of labelling agent and blocking was followed. The lysed protein sample was pipetted onto the membrane and incubated for 12 h at 4 °C on a rocker, then washed twice with provided wash buffer, incubated with detection buffer for 30 min then washed a further three times. Membranes were developed and imaged using an iBright CL750 imaging system. For cryoEM analysis, samples were vitrified (FEI, Vitrobot 2) and images were captured (FEI, Tecnai F20) by technicians at Academia Sinica Cryo-EM facility (Taipei, Taiwan).

2.5. Cell sources, routine culture and hypoxic injury models

BM-MSCs from healthy young donors were purchased from Lonza (PT-2501, lines 38157, 39060 and 39334). Cells were cultured in MSC growth medium (Lonza, PT-3238) and growth supplement (PT-3011) using EV-depleted FBS (Thermo, A2720801). Cells between passage 2 and 5 were used to condition medium for 3 days. Human cardiac microvascular endothelial cells (hCMVECs) were purchased from Lonza (CC-7030) and cultured in Human Large Vessel Endothelial Cell Basal medium (HLVECB; Gibco, M – 200) supplemented with Large Vessel Endothelial Supplement (LVES; Gibco, A1460801). Rodent cardiomyoblasts (H9C2) were purchased from BCRC, Taiwan (60096) and cultured in DMEM (Gibco, 11885-084) with 10% v/v FBS (HyClone, SH30396.03) and 100 IU/ml penicillin, 100 µg/ml streptomycin (P/S; HyClone, SV30010). Human cardiomyocytes (AC16) were purchased from Merck (SCC109) and cultured in DMEM/F12 medium (Gibco, 11330-032) with 10% v/v FBS and 1% P/S solution. All cell cultures were routinely cultured at 5% CO₂ at 37 °C. Prior to hypoxia induction, cells were cultured in EV-depleted FBS (Thermo, A2720801) for 48 h to minimise interference from bovine EVs. Hypoxia was induced *via* dual-

gas incubator, set to 94 % N₂, 5 % CO₂, or by using the Anaeropack system (BD). To induce hypoxia/re-oxygenation injury, cells were returned to 5% CO₂.

2.6. Measurements of cell metabolic activity and apoptosis

To measure cell metabolic activity, 10 µl CCK-8 solution (Boster, AR1160) was added to each well of a 96-well plate and incubated for 2–4 h. The absorbance was then read with a microplate reader at 450 nm. Basal medium, without cells, was used as a blank for all assays; the values of which were subtracted from sample values. We found that SCPL-W alone had a small amount of CCK-8 reagent conversion, likely due to residual enzyme activity in the platelet lysate; this was subtracted as necessary. SCPL-EVs lacked this activity. For flow cytometric analyses, AC16 cells were seeded into 6-well plates (4500 cells/cm²). At ~80% confluency, cells were subjected to 5/6 and 5/8 hypoxia/re-oxygenation injury and apoptosis was analysed using an Annexin-V/PI kit (MedChemExpress, HY-K1073). Floating cells were pooled with adherent cells, detached using TrypLE reagent, then centrifuged at 300×g for 5 min and resuspended in 1 ml of Ca²⁺- and Mg²⁺-free DBPS. Each FACS tube was loaded with 300,000 cells through the filter cap, centrifuged at 300×g for 5 min, then resuspended in 100 µl 1x Binding Buffer. 5 µl of Annexin V-FITC and 5 µl of propidium iodide (PI) was added per tube, then incubated for 15 min at room temperature in the dark. After the incubation period the volume was brought up 400 µl total with binding buffer. All tubes were kept on ice in the dark until measurement. A BD FACSDiscover S8 was used to interrogate the cells using blue laser excitation and detection. At least 5000 events were collected for each sample. The instrument delivered compensated data files based on the initial calibration; gates to define negative and positive populations were identified using unstained cells, single-stained cells, and dead (5% DMSO, 1 h) cells. FlowJo V10.10 and BD CellView V1.2.3 were used to analyse the data. Results were compared by two-way ANOVA vs. the vehicle control group.

2.7. Human cardiac microvessel endothelial cell (hCMVEC) angiogenesis assay

10 µl growth factor-reduced Matrigel (Corning, 354230) was added to 96-well microplates (Ibidi, 89646) and incubated for 30 min at 37 °C to polymerise. hCMVECs were seeded (6.4 × 10³ cells in 70 µl HLVECB medium per well) and immediately treated with SCPL-EV or SCPL-W. HLVECB with LVES supplement was used as the positive control.

2.8. Murine cardiac ischaemia/reperfusion injury model

All animal experiments were conducted in 12-week-old male immunocompetent C57BL6/J mice which were purchased from National Laboratory Animal Center, Taiwan. Their use was approved by Academia Sinica IACUC (Protocol AS16-12-1015). Animals were housed in a specific pathogen free facility, with a 12/12 night/day cycle and *ad libitum* access to food and water. Surgeries were carried out by a licensed veterinarian. To induce myocardial I/R injury, mice were first anaesthetised by inhaled isoflurane/air and supported by intubation and ventilation. The chest was opened by left thoracotomy, the pericardium was carefully displaced, then the left anterior descending coronary artery was ligated with a reversible prolene suture for 45 min, followed by reperfusion. The sham surgery followed the same exact procedures without the coronary artery ligation step. Buprenorphine (0.05 mg/kg, subcutaneous) was given to all mice following surgery. SCPL-EVs (6 µg/g, corresponding to approximately 5 × 10¹⁰ particles measured by NTA), or an equal volume of vehicle control, were given 24 h later by echo-guided percutaneous intracavitary (EGPI) injection, following a previously-described method [41]. This was followed by intravenous (tail vein) administrations at D3, D5 and D7 post-I/R. Twelve batches of SCPL-EVs were isolated, pooled and then concentrated so that the dose

could be administered in a volume of ≤ 80 μL . Cardiac function was measured at D1 and D28 post-I/R injury by echocardiography (Vivid-q ultrasound machine, GE), performed by a technician who was blind to the experimental groups, with mice identified by numerical ear tags. Mice were placed under mild inhaled anaesthesia to obtain a heart rate of ≥ 400 bpm for echocardiography. Mice which died within 24 h of the surgery (i.e. prior to assignment to treatment groups) were excluded from our analysis. In addition, three mice failed to show $\text{EF}\% < 65\%$ at D1 post-I/R surgery and were excluded from the study, again prior to assignment into treatment groups. Before sacrifice, a Millar SPR-839 rodent catheter was inserted into the left ventricle chamber via the right carotid artery. Blood was withdrawn for catheter calibration, then the MVPS system was used to record pressure-volume (PV) loops in the left ventricle for 20 min. Powerlab software was used to calculate dP/dT values based on the last 10 min of recordings. The PV loops were analysed by an independent researcher blind to the experimental groups. Organs were collected for weighing, histological sectioning and staining following standard lab protocols. Heart cross-sections at the mid-infarct area were fixed, paraffin embedded and sectioned, then stained. For alpha-SMA-Cy5, sections were blocked (Biotna, TATS01F), stained for 1 h (Sigma, C6198, 1:200 dilution) at room temperature, washed, then counter-stained with DAPI. Images were captured with a Zeiss Stellaris confocal imaging system. CD31 was stained following standard immunohistochemistry methods using Genetex, GTX20218 at a 1:500 dilution for 1 h at room temperature, followed by secondary antibody for 30 min at room temperature, then 5 s in haematoxylin. Chromogen DAB was developed for 4 min. Reagents used were from Biotna IHC labelling kit (TAHC04D). Quantification was based on analysing three fields per animal from the remote and infarct zones.

2.9. EV labelling and biodistribution assessments

To track *in vivo* EV distribution and uptake, EVs were covalently labelled using an Alexa568 protein label kit (Invitrogen, A201184) and unbound free dye was removed using the provided columns. EV delivery after a single EGPI injection was confirmed 30 min after administration in cardiac I/R-injured mice. The heart, lungs, liver, spleen, kidneys and whole blood were collected and imaged using IVIS Lumina III XMRS (excitation 580 nm, emission 620 nm), operated by a core facility technician. The EV concentration per organ was quantified from a standard curve made from labelled EVs and normalised to the sample weight. Organs from mice injected with unlabelled EVs were used as a blank, present in each frame. For *in vitro* uptake, H/R-injured AC16 cells were incubated with Alexa680-labelled EVs (Invitrogen, A20172) or unlabelled EVs for 30 min, washed 2x with PBS, fixed, stained with DAPI and imaged by Zeiss Stellaris confocal microscopy system. Images were acquired at the same settings to allow direct comparison.

2.10. Murine biocompatibility and coagulation tests

Healthy (without I/R) mice were injected with SCPL-EVs (6 $\mu\text{g/g}$) at D0, D3 and D5 by tail vein. Blood was obtained by cardiac puncture at D7 and immediately mixed 9:1 with 3.2 % citrate anticoagulant (0.105 M). As an additional test, mouse whole blood was spiked with 4 % (v/v) SCPL-EVs and incubated for 30 min to simulate a high intravenous SCPL-EV dose. Plasma was separated by centrifugation ($1,500 \times g$, 10 min). 25 μL plasma was then aliquoted, mixed with 25 μL APTT-S or PT-S reagent (TeClot A0300 or A0230) and pre-warmed to 37°C for 3 min 25 μL CaCl_2 (0.025 M) was then added and PT/aPTT were measured using a Coatron M2 analyser. Other organs were taken, fixed in formalin, embedded in paraffin, and sectioned for histology following standard lab protocols.

2.11. Profiling of SCPL-W and SCPL-EV miRNA content

miRNAs were extracted from SCPL-W or SCPL-EVs using a qEV RNA extraction kit (IZON, RXT01) after spiking the included lysis buffer with

known miRNA standards (Qiagen, 339390). RNase controls were carried out as described earlier. Reverse transcription of RNA was carried out using a miRCURY LNA RT kit (Qiagen, 339340), and a miRCURY LNA SYBR Green PCR kit (Qiagen, 339347) was used for amplification. Successful extraction and amplification of the spike-in miRNAs was confirmed with miRCURY LNA miRNA QC PCR Panel (Qiagen, 339331) in 96-well format using ABI StepOnePlus system. The full human panel I + II V5 panel (Qiagen, 339332) was then run in 384-well format using a Roche Lightcycler 480 instrument. Plates were normalised to one another by using the inter-plate calibration wells built into each plate. miRNAs with cycle threshold (CT) values below 35 were included in the analysis. Sample CT values were normalised against the entire geNorm reference miRNA panel, using the Qiagen Geneglobe (V1.0) service. The DAVID 6.8 tool (<https://david.ncifcrf.gov/summary.jsp>) was used to perform functional annotation clustering of the top 50 miRNAs in SCPL-W and SCPL-EV.

2.12. Profiling of SCPL-W and SCPL-EV protein content

For SDS-PAGE analysis, 10 μg protein was mixed with XT sample buffer (Bio-Rad, 1610791) and heated at 95°C for 5 min. Samples were then loaded into precast NuPAGE 4–12%, 1.0 mm, Bis-Tris, protein gel (Invitrogen, NP0321BOX) with or without a reducing agent (Invitrogen, NP0009). Antioxidant (Invitrogen, NP0005) was added in the running buffer for electrophoresis. Electrophoresis was performed for 30 min at 60V, then 90 min at 125V in running buffer (T-Pro Biotechnology, JB07-G003). Protein gels were visualised using iBright750 (Invitrogen). For proteomic analysis, proteins were isolated as previously described [42]. LC-MS/MS analysis was performed at National Taiwan University proteomics core facility using an Orbitrap Fusion Lumos Tribrid quadrupole-ion trap-Orbitrap mass spectrometer. Peptides were separated on an Ultimate system 3000 nanoLC system, after loading onto Acclaim PepMap NanoLC column with 2 μm particles and 100 Å pores. Mobile phase A was 0.1% formic acid in water, and mobile phase B was 0.1% formic acid in acetonitrile. A gradient from 2% to 40% solvent B was run over 90 min at a flow rate of 300 nL/min. Mass spectrometry analysis was performed in a data-dependent mode with Full-MS (externally calibrated to a mass accuracy of < 5 ppm, and a resolution of 120,000 at $m/z = 200$, AGC target $5e5$, maximum injection time of 50 msec) followed by HCD-MS/MS of the most intense ions in 3 s. High-energy collision activated dissociation (HCD)-MS/MS (resolution of 15,000) was used to fragment multiply charged ions within a 1.4 Da isolation window at a normalised collision energy of 32. The raw MS/MS data were searched against the UniProt knowledgebase reviewed (Swiss-prot) human protein database (downloaded in August 2017) by using the Mascot search algorithm via the Proteome Discoverer (PD) package (version 2.2, Thermo Scientific). Protein identification was supported by at least two unique peptides, a MASCOT score of ≥ 30 , and false discovery rate lower than 0.05. Proteins quantified in at least one replicate were considered as identified proteins, among which proteins quantified in at least two replicates were statistically analysed. Abundance was assumed to be normally distributed within each group, then F-test was applied to see if variance is present between groups. Pooled t-tests were utilised to compute the p-value corresponding to the homogeneous variance; otherwise, Welch's t-tests were applied for the homoscedastic variance. Proteins with $p < 0.05$ and ≥ 2 times fold-change were considered statistically significant. Principal component analysis (PCA) was computed to visualise individual proteomes. Identified proteins of each group were Venn-plotted and common proteins were z-score transformed in abundance, then visualised using a heat map combined with hierarchical clustering (average-linkage and Euclidean distance). DAVID 6.8 was used to perform predictions of biological pathway, molecular function and cellular component.

A semi-quantitative antibody-based membrane (AbCam, ab193655) was used to profile pro- and anti-angiogenic proteins in SCPL-W and SCPL-EVs. The same amount of protein (50 μg) was added to each, as

specified by the manufacturer, and the membrane was stained and developed following the manufacturer protocol, then imaged on an iBright CL650. Positive control spots were used to normalise the samples across each membrane. Protein spot integrated density was measured from multiple exposures using ImageJ following the manufacturer's recommended method.

2.13. Macrophage polarisation

Bone marrow cells were collected from mouse femurs following a published method [43]. Experiments were approved by TMU IACUC (LAC-2020-0529). Cells were differentiated to bone marrow-derived macrophages (BMDMs) using 5 ng/ml CSF-1 (Gibco, PMC2044). After 7 days, >96% of cells were positive for CD11b and >86% were positive for F4/80 (in-house data). BMDMs were incubated with 0.5 µg/µl SCPL-W, SCPL-EV or a matching volume of vehicle control (naïve cells) for 24 h. Lipopolysaccharide (Sigma, SI-L4391) at 0.1 µg/ml or recombinant murine IL-4 (Peprotech, 214-14) at 20 ng/ml were used as controls to induce M1 and M2 phenotypes respectively. RNA was extracted using RNeasy kits (Qiagen, 74106), reverse transcribed with SuperScript IV (Thermo, 18-091-050) and then analysed by RT-qPCR using SYBR green (Thermo, 43-687-08) in a StepOnePlus system. Gene expression was normalised to *Hnnpab*, which had a cycle threshold value of 23–24 in all samples. Log2 fold change vs. naïve vehicle was then calculated. Primers are shown in Supplemental Table 1.

2.14. Data handling

Acquired data was compiled in Excel, Google Sheets or Apple Numbers. Statistical analysis was performed, and graphs were made using GraphPad Prism 9 software. Individual samples are shown in the figures where available, and statistical tests are described in figure legends and results text. Fig. 1A was created using Biorender. Final figure panels were assembled in Affinity Designer.

3. Results

3.1. Characterisation of chromatographic fractions of platelet concentrates

Pooled human PCs were obtained from Taipei Blood Center. SCPL was then generated as shown in Fig. 1A. Whole SCPL (SCPL-W), which contains a mixture of EVs, platelet proteins and remnant serum proteins, was loaded into size exclusion chromatography columns and separated into 30 fractions. The protein content of each fraction was measured by spectroscopy, and the particle size of each fraction was determined by NTA. A plot representing the average of three separate batches showing protein concentrations and particle sizes for each individual fraction is shown in Fig. 1B. The first four fractions contained very few particles or proteins, as expected. From the fifth to tenth fractions, particles of 80–150 nm diameter were detected with the exclusion volume, but the total protein content remained low. After the tenth fraction, much higher concentrations of protein were eluted from the column, peaking at the 17th fraction (32.7 mg/ml). The particle sizes in these fractions were smaller (≤ 50 nm) than the typical EV range. Based on these results, we categorised fractions 7 to 11 as the SCPL-EV fractions and 15 to 19 as the SCPL-protein (SCPL-P) fractions. Those fraction ranges were then pooled together for further characterisation. Measurement of protein concentration by BCA assay (Fig. 1C) confirmed that the protein fraction had significantly lower concentration than SCPL-W, and the SCPL-EV fraction was lower still (148 µg/ml). This demonstrates that size-exclusion chromatography successfully removed >99.5 % of the total protein from SCPL-W. The total particle count of the pooled fractions was measured by NTA, as shown in Fig. 1D. The SCPL-EV fractions had a lower total particle count (4.46×10^{10} /ml) than SCPL-W (1.41×10^{11} /ml), as expected due to dilution during size exclusion chromatography.

SCPL-EV particle counts from four separate batches were highly consistent, with a coefficient of variation of 6.07 %. To visualise particle size distribution of SCPL-EVs more precisely, the pooled fractions were analysed by dynamic light scattering (DLS). Three representative samples of each fraction are shown in Fig. 1E and average particle diameters are shown in Fig. 1F. SCPL-W showed a broad peak containing larger diameter particles (259.3 ± 185.7 nm). On the other hand, the SCPL-P fractions showed multiple peaks, with the most abundant in the 10–30 nm range (average 20.41 ± 6.8 nm). This is likely due to proteins such as albumin and immunoglobulins, possibly aggregated, which are within this size range [44]. On the other hand, the SCPL-EV fractions showed single, narrow, peaks in the 100–200 nm range, with high reproducibility (average 110.6 ± 9.6 nm, coefficient of variation 17.4 %) across four batches. The particle to protein ratio can be used to assess EV purity, since each EV carries a finite amount of surface and cargo protein [45]. The results (Fig. 1G) showed that the SCPL-W and SCPL-P fractions contained a low particle:protein ratio of 3.9 and 8.0×10^6 particles per µg protein respectively, indicating the presence of abundant free proteins. The SCPL-EV fraction showed a significantly higher ratio (3.1×10^8 particles per µg), indicating a higher purity of EVs. Lastly, we extracted DNA or mRNA from the fractions and measured the concentrations (Fig. 1H and I). The DNA concentration was negligible in all preparations, consistent with the fact that platelets are anuclear and contain very low amounts of DNA [46]. Similarly, mRNA content was low, and was not affected by RNase treatment, suggesting that SCPL-related mRNA are part of the EV cargo. These results agree with previous studies showing that the majority of circulating nucleic acids are associated with EVs [47,48]. Taken together, the data show that size-exclusion chromatography suitably removed most free proteins from platelet lysates and produced fractions of uniform 110 nm particles, which were thought to be EVs, as confirmed below.

3.2. Characterisation of isolated SCPL-EVs

To confirm whether the SCPL-EV fraction contained EVs we used a membrane-based human-specific antibody array to detect several EV/exosome protein markers, as shown in Fig. 2A. In SCPL-W well-known tetraspanin markers CD63 and CD81 were both positive, as were the cargo and biogenesis markers TSG101, ALIX, FLOT1 and ANXA5. EpCAM, found in higher levels on epithelial cells, was notably weaker than the other bands. GM130, a Golgi marker, also showed some signal in SCPL-W. The SCPL-EV fractions also showed presence of all EV markers but GM130 was notably weaker, reflecting less cell debris contamination. SCPL-W treated with a commercial EV precipitation reagent (ExoQuick) was used as an additional control, which was positive for all markers. (Supplemental 1A) Next, cryoEM was used to examine the morphology of the SCPL-W and SCPL-EV fractions (Fig. 2B). SCPL-W contained numerous ~100 nm vesicles with bilayer membranes, typical of EVs. However, we also noted many other nano- and micro-scale spherical objects which were not membrane-bound. Based on their morphology, these are likely to be lipoproteins (e.g. chylomicrons, VLDL, LDL and HDL particles) [49]. cryoEM of the SCPL-W samples also showed numerous grainy black spots of approximately 10 nm diameter, present outside of EVs, which are likely to be protein aggregates. Examination of the SCPL-P fractions (Supplemental 1B) showed complete absence of any vesicles or larger particles in all images (50+ fields). On the other hand, the purified, pooled SCPL-EV fractions showed abundant lipid bilayer-bound vesicles in the ~100 nm range (Fig. 2B), and a notably cleaner background, likely due to removal of free proteins. This aligns with the DLS and particle:protein ratios. The average diameter of 80 membrane-bound vesicles from two batches (Fig. 2C) was 105.8 ± 41.04 (±SD) nm based on cryoEM, which was very similar to that measured by DLS (Fig. 1F). This was further confirmed by NTA. Results from three separate batches of SCPL-EVs are shown in Fig. 2D and E. Taken together, the results demonstrate that SCPL-W contains a mixture of proteins and EVs, whereas the SCPL-EV

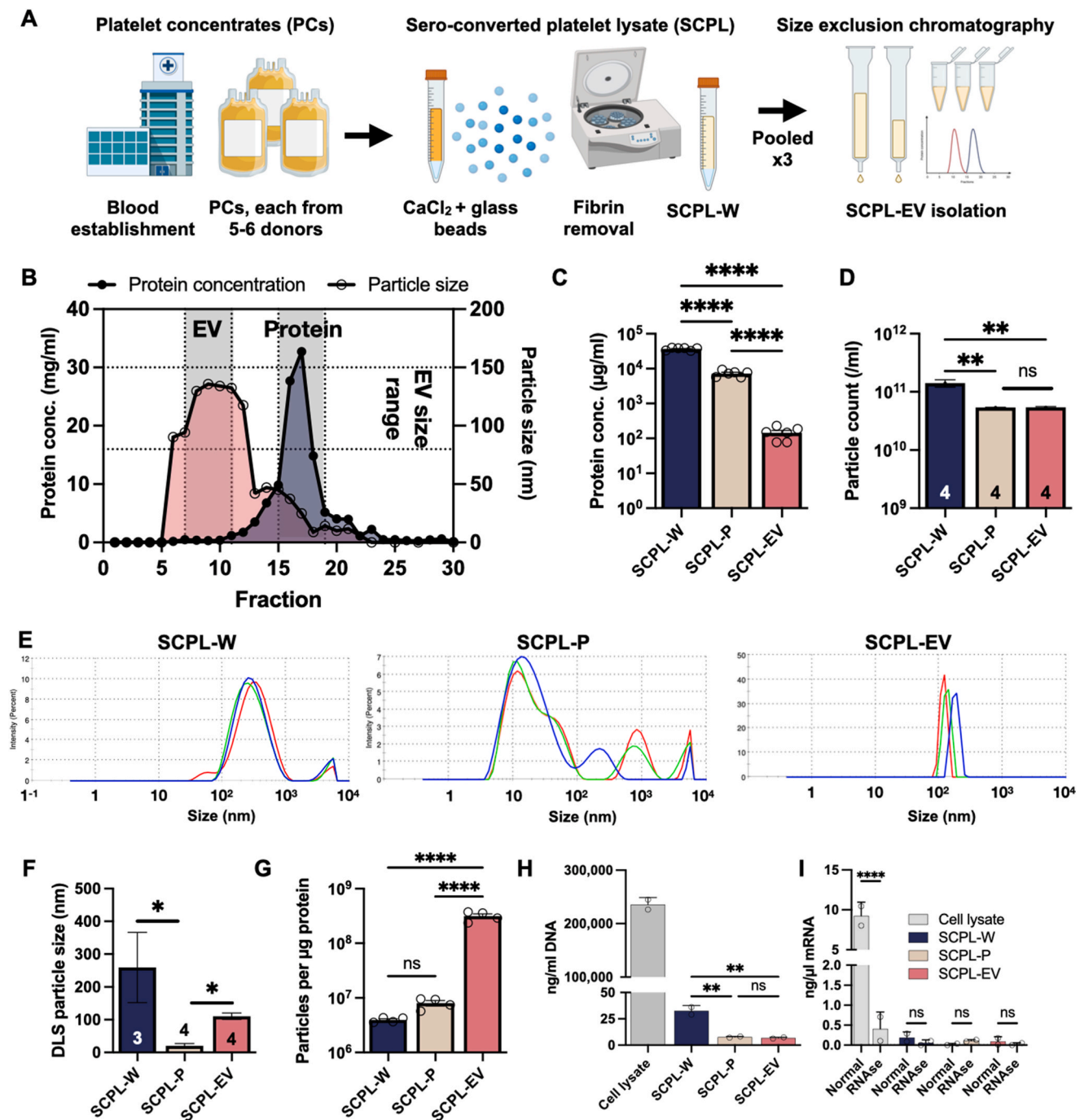


Fig. 1. Generation and separation of sero-converted platelet lysate (SCPL)-derived extracellular vesicles

A. Schematic diagram showing bioprocessing of platelet concentrates into sero-converted platelet lysate (SCPL) and separation of SCPL-EVs by size exclusion chromatography.

B. Dual-axis graph showing protein concentration (left Y axis, blue shaded) and particle size (right Y axis, pink shaded) for each 500 μl fraction (X axis) after size exclusion chromatography. The EV fraction range (7–11) and protein fraction range (15–19) are annotated by dotted lines. The EV size range of 80–150 nm is also annotated on the Y axis. The average of three batches is shown.

C. Protein concentration of whole SCPL (SCPL-W), pooled protein fractions (SCPL-P) and pooled EV fractions (SCPL-EV). The Y axis uses a log₁₀ scale. N = 6 batches.

D. Particle count per ml, determined by nanoparticle tracking analysis (NTA), for SCPL-W, SCPL-P and SCPL-EV. Y axis shows log₁₀ scale. Results of n = 4 batches are shown. Error bars show standard deviation (SD).

E. Dynamic light scattering (DLS) size: intensity plots showing three batches of SCPL-W, SCPL-P and SCPL-EV.

F. Mean particle diameter determined by DLS. The number of batches are shown in the bars. Error bars show SD.

G. Particle to protein ratio for four batches. The Y axis uses a log₁₀ scale.

H. Double stranded DNA (dsDNA) concentration measured in two batches.

I. mRNA concentration measured in two batches, with and without RNase treatment of the SCPL-W. All statistical comparisons in this figure were performed by one-way ANOVA with Tukey's post-test, with all samples compared. * = $P \leq 0.05$, ** = $P \leq 0.01$, **** = $P \leq 0.0001$, ns = not significant ($P > 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

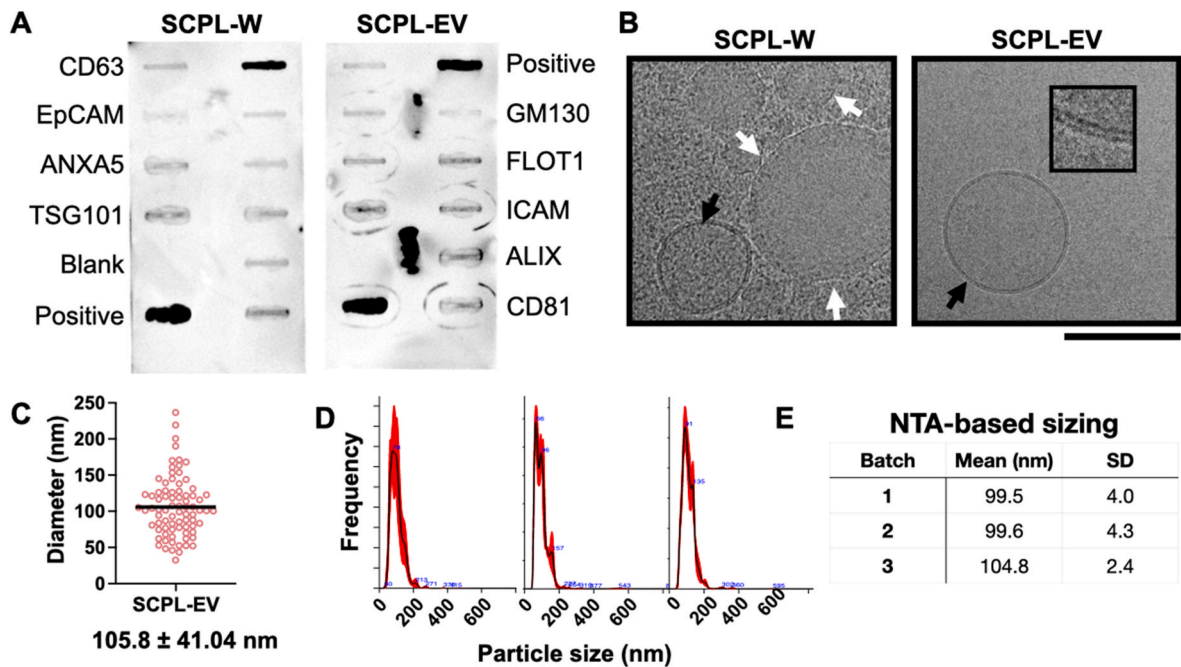


Fig. 2. Confirmation of EV isolation from SCPL. A. Images of human-specific antibody arrays incubated with whole SCPL (SCPL-W) or EV (SCPL-EV) samples. 50 μ g total protein was loaded onto each array. B. Representative cryoEM images of SCPL-W and SCPL-EV fractions. Bi-layered membrane-bound vesicles are labelled with a black arrow. Particles lacking lipid bilayers in the SCPL-W image are shown by white arrows. The SCPL-EV image shows a higher magnification inset to illustrate the bilayer membrane. Scale bar = 100 nm. C. SCPL-EV diameter based on cryoEM images. N = 80 EVs were measured from two batches and the mean $\pm$ SD is shown. D. NTA plots of three batches of SCPL-EVs. E. The mean particle sizes are shown in the table.

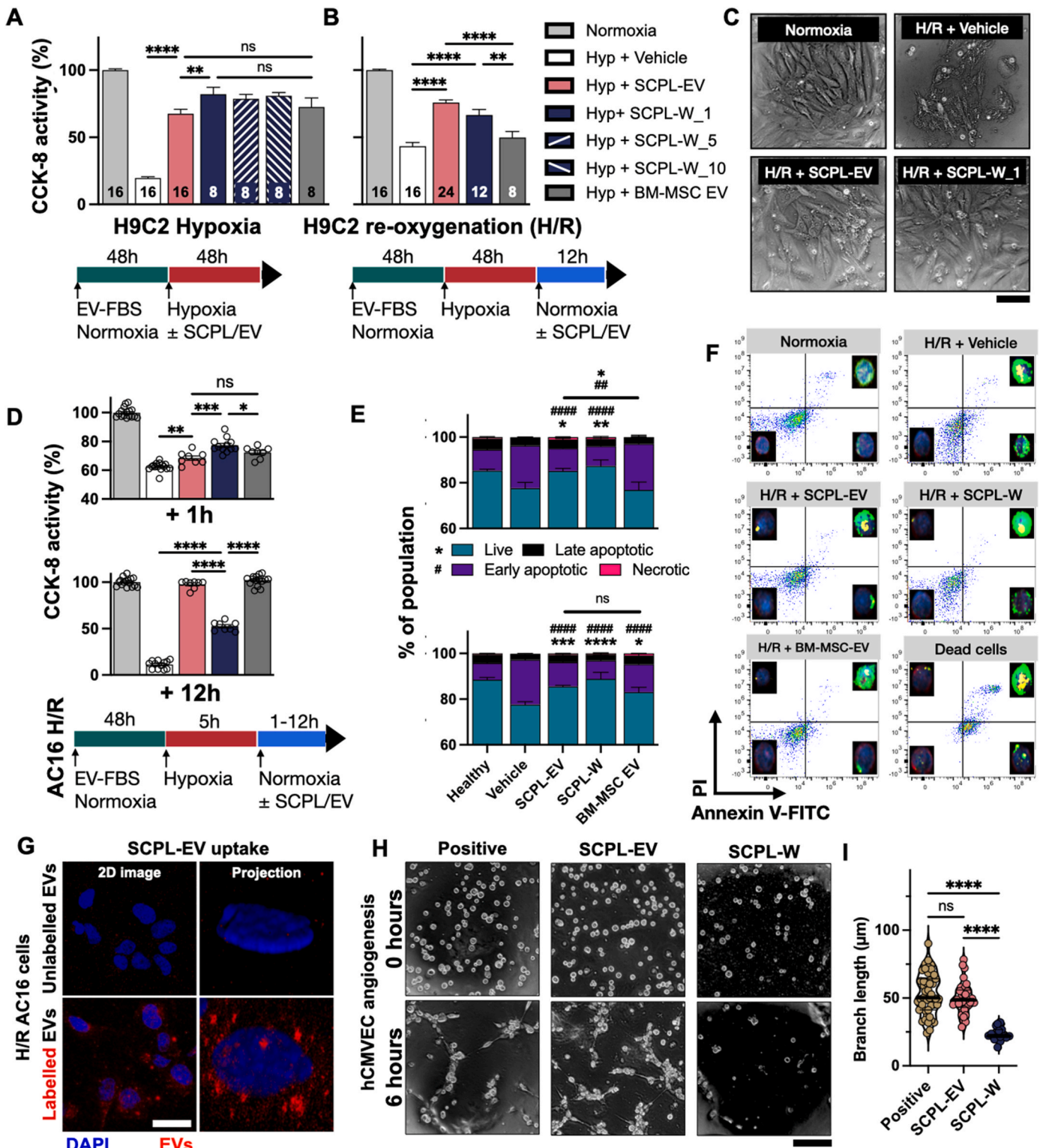
fraction contained a much higher proportion of lipid bilayer-bound vesicles. BM-MSC EVs were also analysed by cryoEM and NTA, showing a high purity of EVs (Supplemental Figs. 1C–D).

3.3. Determining the efficacy of SCPL-EVs using in vitro models of cardiac injury

Next, we sought to determine whether the EVs derived from platelet lysates may have cardioprotective properties. First, we tested the SCPL-W and the SCPL-EV fractions from multiple batches using a rodent cardiomyoblast cell line, which can be used to screen for cardioprotective activity [50]. Two experimental models were used: “hypoxia”, with 48 h hypoxia in the presence of treatments, and “hypoxia/re-oxygenation” (H/R), with 48 h hypoxia, followed by restoration of oxygen and addition of each experimental treatment. The former assay assesses hypoxia-protective effects, whereas the H/R model mimics a clinical situation where blood flow is restored to the injured myocardium and measures the ability of a treatment to acutely protect hypoxic-injured CMs from re-oxygenation injury [21,51,52]. Bone marrow MSC EVs (Supplemental Fig. 1C) were included as a known therapeutic EV for comparison [53]. The results (Fig. 3A) showed that hypoxia significantly lowered H9C2 CCK-8 activity (to $19.7 \pm 0.973\%$, relative to normoxia), as expected. SCPL-W, at a concentration of 1% (v/v), corresponding to 350 μ g/ml of protein, had strongly protective effects ($82.2 \pm 5.0\%$) but further increases in SCPL-W dose to 5% or 10% had no additional effect (78.8 ± 3.3 and $81.0 \pm 2.4\%$ respectively). Therefore, 1% SCPL was used for all future experiments. SCPL-EVs, also given at 1% (v/v) provided significant protection ($67.7 \pm 3.2\%$, $P \leq 0.0001$ vs. vehicle), despite their far lower overall protein concentration. BM-MSC EVs, provided at the same protein concentration as SCPL-EVs, also provided significant protection ($72.7 \pm 2.4\%$), equal to SCPL-EVs ($P = 0.87$). In the H9C2 H/R assay (Fig. 3B), the vehicle control-treated H9C2 cells showed the lowest viability ($43.3 \pm 2.8\%$),

as expected. Both SCPL-EVs ($75.9 \pm 2.0\%$) and SCPL-W ($66.7 \pm 4.1\%$) offered significant protection from re-oxygenation injury, as did BM-MSC EVs ($49.9 \pm 1.6\%$). In this assay, SCPL-EVs had a protective effect equal to SCPL-W ($P = 0.092$), and significantly higher than that of BM-MSC EVs ($P = 0.007$). The viability assay results were supported by the morphology of the cells, shown in Fig. 3C. H/R-injured cells showed disrupted morphology including cytoplasmic vacuoles and membrane fragmentation whereas the SCPL-W and SCPL-EV treated cells appeared more like the normoxia group.

We carried out further testing using human cardiomyocytes (AC16 cells) which are much more sensitive to oxygen deprivation than H9C2 cells. Therefore, the hypoxia duration was shortened to 5 h, and CCK-8 activity was measured at 1 h and at 12 h after re-oxygenation to assess early cardioprotection. The results (Fig. 3D) showed decreased CCK-8 activity after injury ($62.4 \pm 0.7\%$) and protective effects of SCPL-EVs ($68.4 \pm 1.5\%$, $P = 0.03$), SCPL-W ($77.6 \pm 2.0\%$, $P \leq 0.0001$) and BM-MSC EVs ($72.3 \pm 1.4\%$, $P \leq 0.0001$ vs. vehicle, $P 0.23$ vs. SCPL-EV) after 1 h. This shows that all the EV treatments offer acute cardioprotection following re-oxygenation injury. Examining the later time point after re-oxygenation showed that vehicle-treated cell viability continued to reduce over time ($11.3\% \pm 1.2\%$). SCPL-W treatment offered a good degree of cell protection ($52.5 \pm 1.8\%$) but cardiomyocytes treated with SCPL-EVs and BM-MSC EVs had the highest viabilities ($97.7 \pm 1.5\%$ and $102.3 \pm 1.4\%$ respectively, both $P \leq 0.0001$ vs. vehicle control). These data agree with H9C2 results, showing that the SCPL-EVs were superior to SCPL-W at protecting from re-oxygenation injury. The results also indicate that SCPL-EVs have cardioprotective effects equal to BM-MSC EVs ($P = 0.15$ for AC16 5/12). Taken together, the results show that SCPL-EVs can protect both rodent and human cardiomyocytes from hypoxia and oxidative injury and were consistent across several individual batches. To validate these results, apoptosis and necrosis were quantified by flow cytometry using Annexin-V-FITC and PI staining using a milder H/R injury model to



(caption on next page)

Fig. 3. Assessments of cardiomyocyte protection and cardiac endothelial cell angiogenesis

- A. CCK-8 activity of H9C2 cells after 48h hypoxia (1% O₂) with vehicle, BM-MSC EVs, SCPL-EVs or SCPL-W provided at 1%, 5% or 10% (v/v). Hypoxia samples were compared by one-way ANOVA with Tukey's post-test. N ≥ 3 batches of SCPL-EVs were used, with replicates totalling the number indicated in each bar.
- B. CCK-8 activity after 48/12 hypoxia/re-oxygenation (H/R) assay. Samples were compared by one-way ANOVA with Tukey's post-test.
- C. Representative light microscope images of H9C2 cells under each condition. Scale bar 30 μm.
- D. Human cardiomyocyte (AC16) viability assay following a 5/1 and 5/12H/R injury model. Hypoxia samples were compared by one-way ANOVA with Tukey's post-test.
- E. Quantification of live (FITC⁻/PI⁻), early apoptotic (FITC⁺/PI⁻), late apoptotic (FITC⁺/PI⁺) and necrotic (FITC⁻/PI⁺) cells for 5/6 (upper) and 5/8H/R (lower) models. Samples (n = 3 per group) were compared by two-way ANOVA against the vehicle group with * indicating comparisons of live cells, and # comparing early apoptosis.
- F. Example plots from flow cytometric apoptosis analysis of 5/8H/R AC16 cells. The X axis shows Annexin-V-FITC and the Y axis shows PI. A representative cell image is shown for each quadrant.
- G. Images of H/R-injured AC16 cells incubated with unlabelled SCPL-EVs or Alexa680-labelled SCPL-EVs (red) for 30 min. Nuclei were labelled with DAPI (blue). Scale bar 20 μm.
- H. Images of human cardiac micro vessel endothelial cells (hCMVECs) at 0 h and 6 h after addition of positive control (endothelial growth factor supplement), SCPL-W, or SCPL-EVs. Scale bar 100 μm.
- I. Quantification of hCMVEC branch length. Samples were compared by one-way nested ANOVA with Tukey's post-test. * = P ≤ 0.05, ** = P ≤ 0.01 and *** = P ≤ 0.001. ns = not significant (P > 0.05). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

avoid excessive cell death. The results (Fig. 3E and F) showed that hypoxic cells were reduced to 76.0 % live, 20.8 % early apoptotic and 3.2 % late apoptotic at 6 h after re-oxygenation. However, AC16 cells incubated with SCPL-W or SCPL-EVs both showed significantly higher live cell populations (85.2 and 87.3 % respectively), with corresponding reductions in apoptosis. BM MSC-EVs had no significant effect at this timepoint (76.7%, P = 0.9 vs. vehicle control). At 8 h following reoxygenation both SCPL-EV and SCPL-W treated cells had a higher live cell percentage (85.5 % and 88.8% respectively, both P < 0.0001 vs. vehicle). BM MSC-EVs also increased live cells (83.1%, P = 0.014 vs. vehicle), as expected. These results again show that SCPL-EVs had efficacy similar, or superior to BM MSC-EVs which are a widely researched cardioprotective EV.

Lastly, to confirm SCPL-EV uptake into injured cardiomyocytes, we incubated labelled EVs with H/R-injured AC16 cells for 30 min, washed thoroughly, and captured Z-stacked images. The results (Fig. 3G) showed EVs inside the CM cytoplasm, clustered around the nucleus. Use of a covalent protein-bound dye and removal of unbound dye ensures that this is specific uptake of the EVs rather than transfer of the dye. This demonstrates that SCPL-EVs could be rapidly taken up into hypoxic-injured cardiomyocytes.

3.4. Assessing SCPL-EV stimulation of cardiac microvessel endothelial cell angiogenesis

For therapy of myocardial infarction, promotion of cardiac angiogenesis is beneficial to re-establish tissue perfusion and support tissue recovery [4,5]. Previous studies have shown that a distinct platelet lysate can protect corneal endothelial cells from stress and enhance repair *in vitro* [54]. However, since corneal endothelial cells are biologically distinct from cardiac endothelial cells, and our PC processing methodology was different, we performed a more representative assay using primary human cardiac microvessel endothelial cells (hCMVECs). However, it should also be noted that these cells have less tube formation potential than many other endothelial cells in *in vitro* assays. The results (Fig. 3H and I) showed that SCPL-EVs induced hCMVEC branching and elongation to the same degree as the LVEs-supplemented positive control (P = 0.44). Interestingly, SCPL-W showed no positive effect of tube formation on hCMVECs. Since SCPL-W also contains SCPL-EVs, we speculated that this result may be due to the presence of anti-angiogenic or otherwise interfering proteins in the SCPL-W [42].

To evaluate this further, we examined multiple proteins related to angiogenesis using an antibody-based array, shown in Supplemental Fig. 2. The results showed that both SCPL-W and SCPL-EVs contained several known cardioprotective factors, including EGF, IGF-1 and angiopoietin-1 but SCPL-EV fractions contained a higher proportion of angiopoietin-1 and EGF. SCPL-W also contained a significantly higher

proportion of anti-angiogenic proteins plasminogen and endostatin, as well as more of the cytokine RANTES/CCL-5. These differences may help to interpret the observed pro-angiogenic effects of SCPL-EVs compared to SCPL-W.

3.5. Analysis of SCPL-W and SCPL-EV protein composition

Proteomics can be used to profile the makeup of proteins in an EV population [55]. Since size exclusion chromatography dramatically reduced the total protein concentration of SCPL-W and enriched EVs (Fig. 1) and the SCPL-EV and SCPL-W fractions were functionally different, we sought to investigate their overall protein makeup. To build an overall profile of the protein composition, SCPL-W and SCPL-EV proteins were run using SDS-PAGE under reducing and non-reducing conditions. Under reducing conditions, the SCPL-EV group showed a similar profile, but with different proportions of proteins. SCPL-EV showed a lighter band at ~65 kDa (likely to be albumin under both conditions) and stronger bands at 180 kDa compared to the SCPL-W group. Proteomic analysis by LC/MS identified a total of 323 proteins which met our criteria for high confidence identification and were included in our subsequent analyses. Comparing SCPL-W to SCPL-EV proteomic results by principal component analysis (Fig. 4B) and Pearson correlation (Supplemental Fig. 3A) showed that there was good consistency between different batches, and the overall profiles of SCPL-W and SCPL-EV were statistically distinguishable from one another. SCPL-W had a higher average correlation coefficient (0.56) than SCPL-EVs (0.212). This may occur because SCPL-EV protein concentration is more than 100x lower than that of SCPL-W, and some low abundance proteins may not have been detected in all three batches. However, when comparing the high confidence detected proteins, SCPL-W and SCPL-EV had a considerable degree of overlap (282/323 proteins), as shown by Venn diagram (Fig. 4C). This indicates that both SCPL-W and SCPL-EV had similar protein contents, but in altered proportions. 39 proteins were detected exclusively in the SCPL-EV group, including well-known EV markers CD9 and ITGB3 [55]. These proteins must also be present in SCPL-W; thus it is likely that removal of other proteins allowed for detection of lower abundance proteins in SCPL-EVs by proteomics.

Comparison of the protein content (volcano plot, Fig. 4D) shows that 75 proteins were significantly higher, and 27 proteins were significantly lower in SCPL-EV than SCPL-W. The significantly higher proteins included known EV markers Hsp70 (UniProt P11021, 3.3-fold higher than SCPL-W), RAP1B (P61224, 5.9-fold higher) and LGALS3BP (Q08380, 20.2-fold higher), again supporting enrichment of EVs in this fraction [55]. The 27 significantly lower abundance proteins included several members of the SERPIN family (SERPINA4, A6, A10, C1, F2) complement system proteins (C8A, C8B, CFB, C8G, C3, C5, C4A)

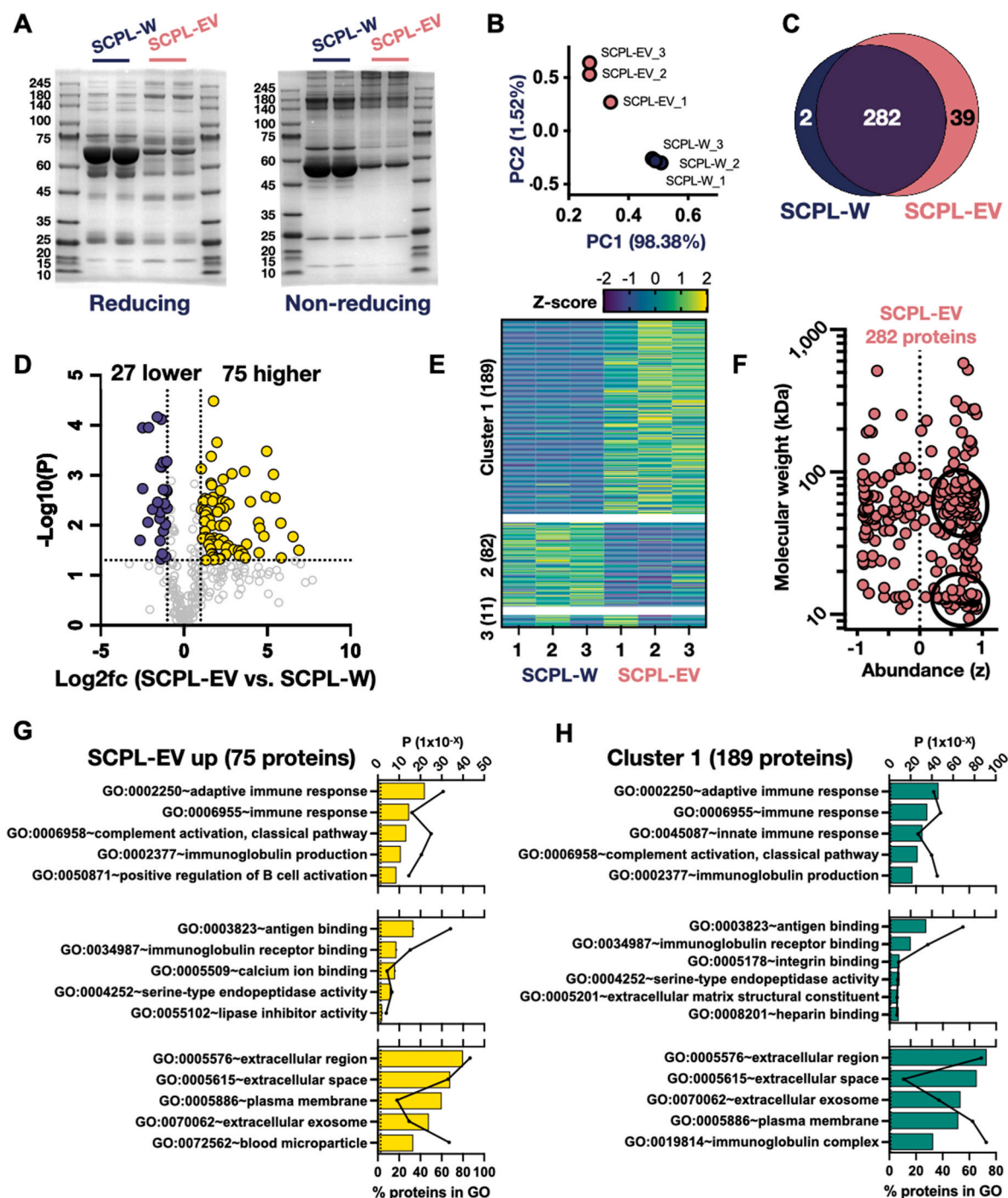


Fig. 4. Proteomic analysis of SCPL-W and SCPL-EV

A. SDS-PAGE of SCPL-W and SCPL-EV run under reducing (left) and non-reducing (right) conditions. The left lane shows the ladder, as annotated. 10 μ g protein was loaded into each lane.

B. Principal components analysis (PCA) of three SCPL-W and three SCPL-EV proteomes.

C. Venn diagram showing proteins identified exclusively in SCPL-W or SCPL-EV, or present in both samples.

D. Volcano plot of 282 proteins detected in both SCPL-EV and SCPL-W. The X axis shows log₂-fold changes of SCPL-EV proteins compared to SCPL-W. Dotted lines on the X axis show fold changes of -2 and +2, and the dotted line on the Y axis indicates $P = 0.05$. Significant increases are shown in yellow, significant decreases are shown in purple, and those with no change (ns) are in grey.

E. Heatmap of 282 proteins, divided into three clusters identified by hierarchical analysis. The number of proteins in each cluster is shown to the left.

F. Plot of SCPL-EV protein molecular weight (Y axis, log₁₀ scale) against mean relative abundance compared to SCPL-W (z-score, X axis). Two circles identify enriched groups of proteins at 10–20 kDa and 60–90 kDa.

G. Graphs showing biological pathway (BP, upper), molecular function (MF, middle) and cellular component (CC, lower) for 75 proteins which were in higher abundance in SCPL-EVs. The lower X axis shows the percentage of proteins in each ontology, represented by the bars. The upper X axis shows the P value (Benjamini FDR, shown as $\times 10^{-X}$), represented by the line.

H. BP, MF and CC analysis of cluster 1 (189 proteins) which was enriched in SCPL-EVs. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

coagulation-related proteins (FGA, F12) and carrier proteins (TF, ALB, TTR, GC). This demonstrates that size exclusion chromatography removed several abundant contaminating proteins. The anti-angiogenic protein CXCL4 (platelet Factor 4) was reduced (log2fc -0.57) in SCPL-EV compared to SCPL-W which may also explain the differences we observed in angiogenesis stimulation [42].

From the 282 common proteins, hierarchical clustering (Euclidean method) revealed three distinct groups, shown in the heatmap (Fig. 4E). Cluster 1 (189 proteins) comprised proteins present in higher abundance in SCPL-EVs, whereas cluster 2 (82 proteins) was higher in SCPL-W. Cluster 3 (11 proteins) showed no particular trend. We noticed that SCPL-EV contained more low molecular weight proteins, as shown in a plot of mean relative abundance (Z score) of SCPL-EV proteins against

molecular weight (Fig. 4F). The results showed two apparent groupings which were enriched in the SCPL-EV samples: at 10–20 kDa and 50–90 kDa. It has also been previously shown that low MW proteins are enriched in EV preparations [39]. Many of the smaller MW proteins in SCPL-EVs were immunoglobulin components such as heavy chain variable regions (various forms of IGHV1, IGHV3, IGLV etc.) and some apolipoproteins (APOC3, APOC2, APOM). It is unlikely that these were freely suspended protein impurities, since neither NTA, DLS nor cryoEM indicated high concentrations of free proteins in solution. Thus, we speculate that these proteins could be contained within the SCPL-EVs, or could be part of their surface corona. Platelets are well-known to possess a canalicular system which serves as two-way communication between blood components and platelet granules, and can internalise small

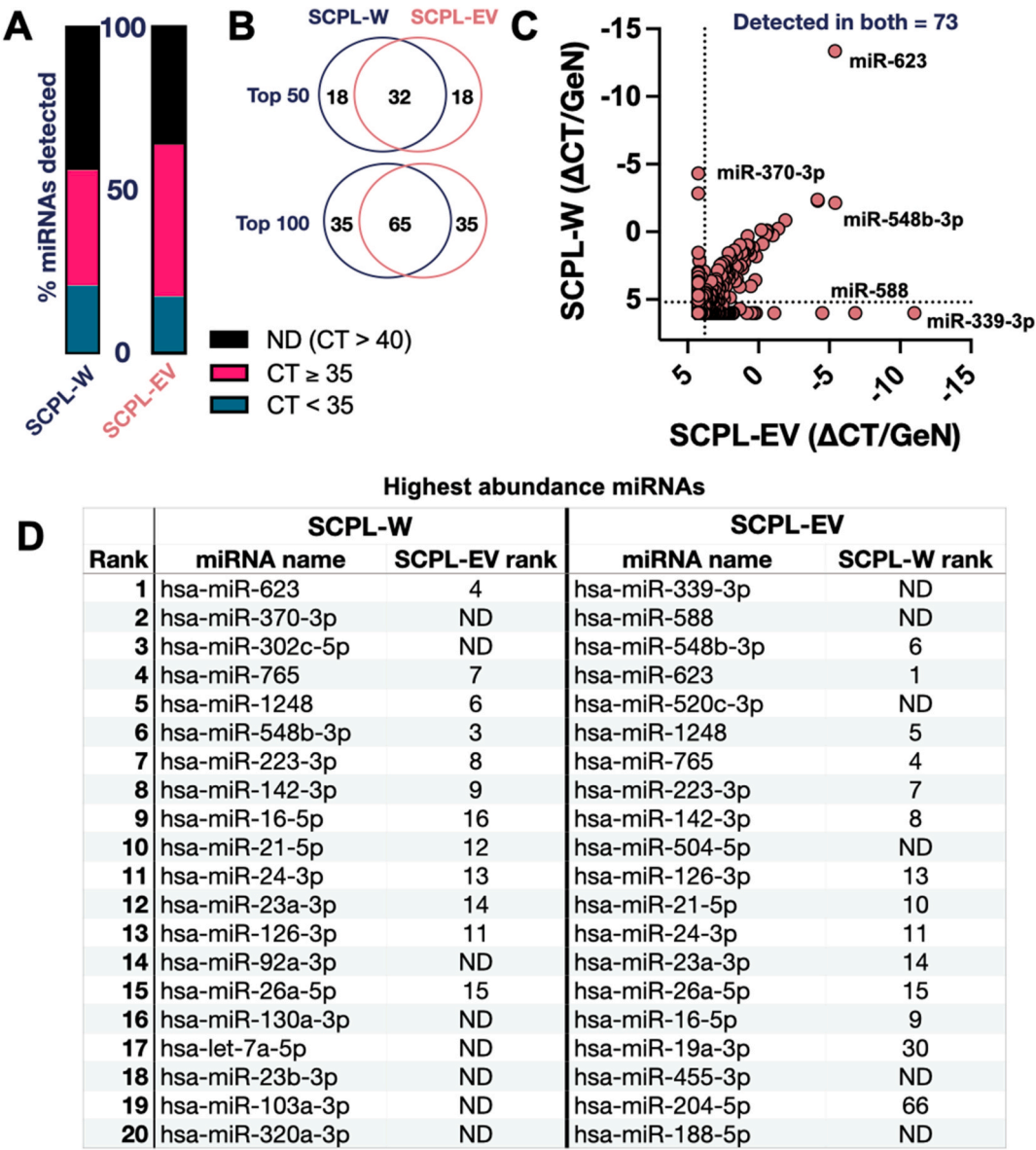


Fig. 5. miRNA analysis of SCPL-W and SCPL-EV

A. Percentage of miRNAs detected in each sample type. Two batches were analysed separately. ND = not detected (CT > 40). miRNAs with CT values of <35 were included in subsequent analyses.

B. Venn diagrams of the top 50 and top 100 miRNAs in SCPL-EV and SCPL-W samples.

C. Scatter graph showing mean miRNA expression levels (normalised to GeNorm (GeN) reference) for SCPL-EV (X axis) and SCPL-W (Y-axis). Selected miRNAs are annotated. The dotted line on each axis shows the cut-off for detection, thus miRNAs in the lower right quadrant were detected only in SCPL-EV and upper left quadrant were detected only in SCPL-W.

D. Table showing top 20 highest abundance miRNAs in SCPL-W (left) and SCPL-EV (right). ND indicates that the miRNA had a CT value of ≥35 and was considered as not detected.

proteins or nanoscale particles [56]. In addition, platelets are known to release different populations of EVs with distinct protein compositions depending upon the triggering factor, consistent with our data [57].

Next, we considered the function of the abundant SCPL-EV proteins. First, we profiled the 75 proteins detected in higher abundance in SCPL-EVs (Fig. 4G). These showed biological pathways and molecular functions related to immune responses, calcium binding and enzyme activity. In terms of cellular component, the proteins were predominantly associated with extracellular origins, which was not surprising. 48% of the proteins were in the “GO:0070062, extracellular exosome” group ($P = 2.4 \times 10^{-15}$). Analysis of cluster 1 (Fig. 4H) showed similar results, with the addition of integrin binding and heparin binding molecular functions. This is of interest because many growth factors such as VEGF and HGF bind to heparin. Analysis of the 39 SCPL-EV exclusive proteins is shown in Supplemental Fig. 3B.

Although the proteomic results confirmed the presence of several EV markers such as CD9, ITGB3 and HSP70 in SCPL-EVs, we noted that our overall dataset had relatively little (~3%) overlap with published proteomic EV datasets in Exocarta and Vesiclepedia (Supplemental 3C-D). This is likely due to relatively low number of P-EV studies (1.8% and 1.4% respectively) contained in those two databases. Comparing our data against published proteomic datasets for platelet lysates (Supplemental Fig. 3E) showed much higher concordance (81.4% overlap with Widyaningrum and colleagues and 69% with Gomes and colleagues) [54,58].

3.6. Analysis of SCPL-W and SCPL-EV miRNA content

microRNAs are known as powerful determinants of EV bioactivity and cardioprotection [59–62]. Therefore, we used a qPCR-based miRNA array to profile >750 known miRNAs in two batches of the SCPL-W and SCPL-EVs. Quality control analyses showed the expected expression of spike-in miRNAs and stably-expressed housekeeping miRNAs. SCPL-EV samples derived from RNase-treated SCPL-W did not show any degradation of housekeeping miRNAs. On the contrary, treating lysed SCPL-EV miRNA extracts with RNase eliminated all miRNAs. This demonstrates that the miRNAs were protected inside EVs. (Supplemental Fig. 4A). The results of the full panel showed that both untreated SCPL-W and SCPL-EVs contained relatively few high abundance miRNAs, as indicated in Fig. 5A. miRNAs with average CT values of less than 35 were included in our analyses, which was approximately 20 % of the total number of miRNAs for both samples. Plotting the normalised expression of all miRNAs for the two SCPL-W samples gave an R^2 value of 0.8010, and SCPL-EVs gave an R^2 value of 0.6935 showing good correlation between two independent batches. The R^2 value of SCPL-W vs SCPL-EV is 0.3012, indicating different miRNA cargo makeup. Venn diagrams (Fig. 5B) showed a ~65% overlap in the top 50 and top 100 miRNAs in SCPL-EVs and SCPL-W. Comparing the expression of each miRNA using a scatter plot (Fig. 5C) showed that there was correlation between SCPL-W and SCPL-EV miRNA content, with 73 miRNAs detected in both SCPL-W and SCPL-EVs. Highly expressed miRNAs in both samples included miR-623, miR-548b and miR-765. However, miR-339, miR-588 and miR-520c were only found in SCPL-EVs, and miR-370 and miR-302c were only found in SCPL-W, suggesting enrichment and depletion of certain miRNAs in the EV population compared to the whole platelet lysate. A table of the top 25 miRNAs in SCPL-W and SCPL-EVs is shown in Fig. 5D. Those with a CT value of ≥ 35 are indicated by ND. To give insights into likely biological functions, additional analyses were carried out using the DAVID functional annotation tool. The results (Supplemental Table 2) found that SCPL-EVs were significantly enriched in miRNA relating to positive regulation of smooth muscle cell proliferation (GO:1904707) and migration (GO:1904754), angiogenesis (GO:1903589, GO:0090050, GO:0035924), the ERK1/2 cascade (GO:0070374) and others. The same analysis of SCPL-W showed similar results, but also enrichment of miRNAs relating to negative regulation of angiogenesis (GO:0016525, GO:1903588) and MMP secretion (GO:1904465) and positive regulation of apoptosis

(GO:0043065) (Supplemental Table 3). Target gene prediction for SCPL-EV enriched miRNAs was carried out using Geneglobe, attached as a Supplemental File. Lastly, we compared the miRNA profile of SCPL-EVs against a published human serum EV dataset, and an in-house miRNA profile of human stem cell derived EVs (Supplemental Fig. 4B) [22]. The results showed that SCPL-EVs miRNA were more similar to serum EVs than stem cell EVs, but still contained many miRNAs which were not found in either population. This again highlights the diversity of EV cargo depending on their cellular origin. The functions of abundant SCPL-EV miRNAs were investigated by literature search; additional information is available in Supplemental Table 4. The results indicated that many of the detected miRNAs have known cardioprotective properties, including miR-765, miR-223-3p, miR-142, miR-21-5p and miR-23a. Many of the identified miRNAs also have known anti-inflammatory activities and were associated with macrophage polarisation. We also noted that the protein array (Supplemental Fig. 2) showed the presence of proteins such as IL-1, IL-10, RANTES/CCL5 and TNF- α in both samples. Therefore, we assessed whether SCPL-W or SCPL-EVs would have polarising effects on naïve macrophages, using gene expression to detect M1 and M2 phenotypic markers, as shown in Supplemental Fig. 5. LPS strongly up-regulated pro-inflammatory markers including *Il1b*, *Il6*, *Tnfa* as expected, and IL-4 increased M2 markers *Arg1* and *Retnla*. Interestingly, we found that both SCPL-W and SCPL-EVs induced a strong M2 signature, significantly increasing *Il10* and *Retnla* expression by more than 40-fold. SCPL-EVs also increased M2 marker *Arg1* by 5-fold. Together, this indicates that SCPL-W and SCPL-EVs have direct anti-inflammatory activities on macrophages [63].

3.7. Investigating uptake of SCPL-EVs by I/R-injured hearts

Although SCPL-W did show some cardioprotective effects in *in vitro* hypoxia protection assays, SCPL-EVs had superior efficacy in the H/R assays, particularly given their lower total protein concentration. In addition, SCPL-W failed to stimulate cardiac endothelial cell tube formation, whereas SCPL-EVs promoted angiogenesis (Fig. 3H and I). This may be explained by the higher proportion of anti-angiogenic miRNA contained in SCPL-W (Supplemental Table 3). Lastly, SCPL-W also contained more coagulation factors and complement proteins, which were greatly reduced in SCPL-EVs (Fig. 4). Therefore, we elected to examine the effects of only SCPL-EVs in animal models. First, we measured SCPL-EV delivery and uptake by the I/R-injured myocardium following echo-guided percutaneous intracavitary (EGPI) administration. To accomplish this, we carried out a pilot experiment, as shown in Fig. 6A. 45 min coronary artery occlusion followed by reperfusion produces a moderate-to-severe degree of injury and causes a mild enhanced permeability and retention effect at the injury site [64]. IVIS analysis (Fig. 6B–D) showed clear uptake of labelled EVs by the I/R-injured heart 30 min following administration. Examination of other major organs (Fig. 6D) revealed EV signal in the liver, with minimal signal in the spleen, kidneys or lungs. Quantifying EV delivery by percentage of the original injected dose (Fig. 6B), using a standard curve (Supplemental Fig. 6A–B), revealed that approximately 0.8 % of the injected EVs were retained by the heart. This is within the typical expected range, given that the heart is known to display exceptionally poor uptake and retention of donor therapeutic cells, EVs, nanomedicines and other therapeutics [64–66]. However, we noted that approximately six percent of the injected dose remained in the blood 30 min after injection, which is considerably higher than EVs from other sources [66,67]. Normalising the delivered EV dose to organ weight (or blood volume), (Fig. 6C), showed that the heart had the highest relative retention of EVs.

3.8. Examining effects of SCPL-EV on coagulation and organ morphology

Since SCPL-EV isolates still contain immunoglobulins and coagulation factors (Fig. 4), we aimed to determine whether there were any

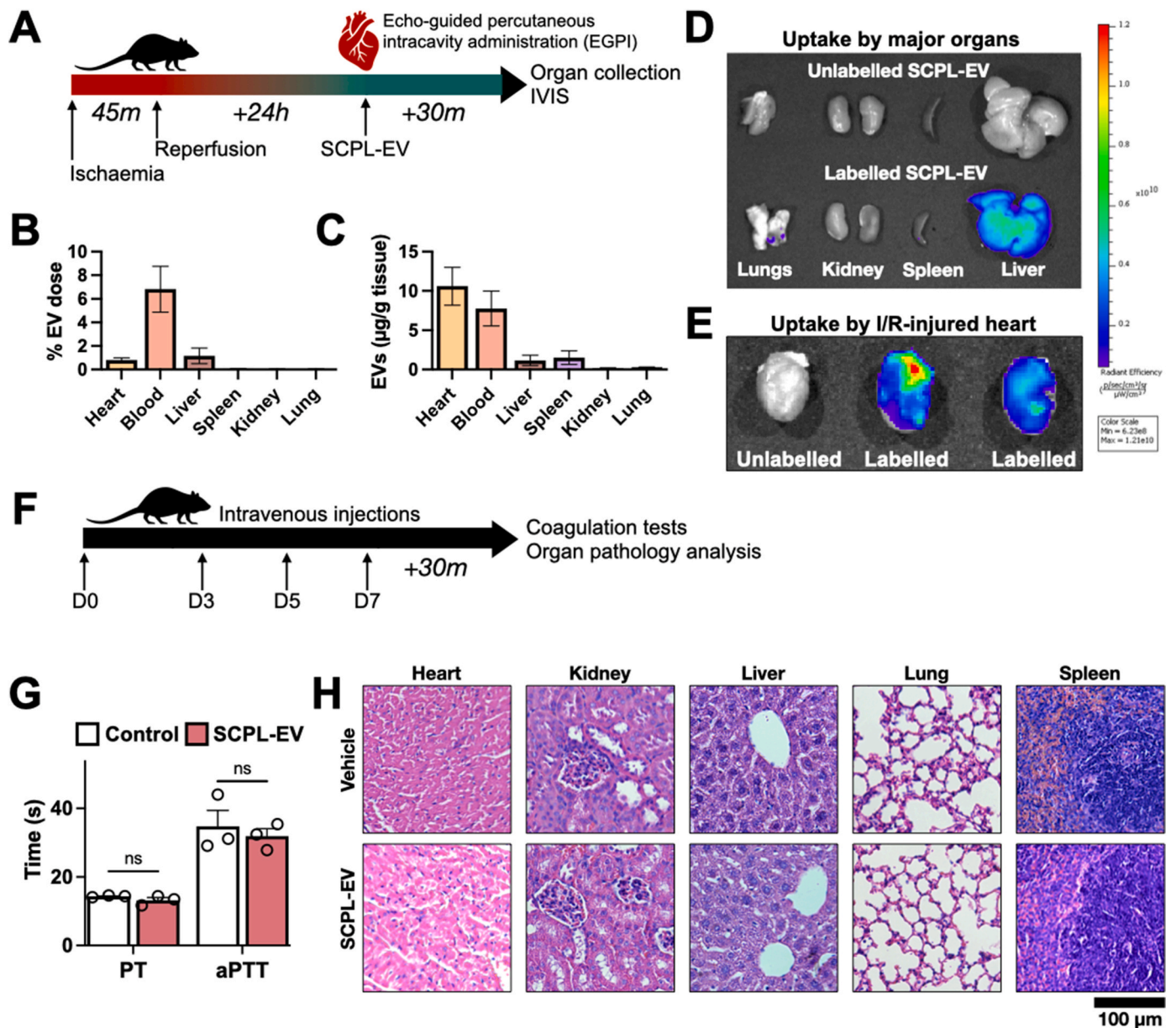


Fig. 6. Cardiac uptake and *in vivo* biocompatibility of SCPL-EVs

A. Schematic diagram of cardiac ischaemia/reperfusion (I/R) injury and SCPL-EV administration by echo-guided percutaneous intracavity (EGPI) administration.

B. Percentage of injected EV dose detected in major body organs and blood.

C. Quantified EV concentration normalised to tissue weight and blood volume.

D. Example IVIS images of lungs, kidney, spleen and liver from cardiac I/R-injured mice injected with unlabelled (upper) or labelled (lower) SCPL-EVs. The colour scale bar for both images is shown to the right.

E. Example IVIS images of I/R-injured mouse hearts 30 min after injection of unlabelled (left) or labelled (middle, right) SCPL-EVs.

F. Schematic diagram of experimental design to assess SCPL-EV safety.

G. Prothrombin time (PT) and activated partial thromboplastin time (aPTT) of mouse plasma after vehicle control or SCPL-EVs. Samples were compared by unpaired *t*-test. ns = not significant ($P > 0.05$).

H. Representative H&E-stained histology images of heart, kidney, liver, lung and spleen from mice receiving vehicle control or SCPL-EVs. Scale bar 100 µm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

overt signs of toxicity, thrombogenicity or immune response following repeated doses of SCPL-EVs. After four intravenous doses over seven days (Fig. 6F) there was no significant change in mouse body weight (Supplemental Fig. 6C). We also measured prothrombin time (PT) and activated partial thromboplastin time (aPTT) of control and SCPL-EV mouse plasma at D7, which were all within the normal range [68]. Additional mouse blood samples, spiked with 6 µg/ml SCPL-EVs 30 min prior to the test to simulate a high intravenous dose, showed no significant difference (Fig. 6G). This indicates that the SCPL-EVs did not have

obvious pro-coagulant activities. Gross anatomical appearance and histology of heart, kidney glomeruli and renal tubules, liver lobules, lung alveoli and spleen red and white pulp were all normal (Fig. 6H).

3.9. Efficacy of SCPL-EVs as a therapeutic following cardiac ischaemia/reperfusion injury

Based on these positive results, we examined the efficacy of SCPL-EVs in a full *in vivo* experimental model of I/R injury, as shown in

Fig. 7A. Mice received 45 min cardiac I/R injury followed by EGPI administration of SCPL-EVs at 24h, then intravenous injections at D3, D5 and D7. Echocardiography showed that all I/R-injured mice showed a significant reduction in cardiac left ventricle ejection fraction % (EF%) at D1, prior to any treatment (**Fig. 7B**). There was no significant difference between the two I/R-injured groups (51.8 ± 2.1 vs. 51.6 ± 1.9 %, $P = 0.99$), but both were significantly lower ($P \leq 0.0001$) than the sham-operated mice. Sham-operated mice showed a small reduction in average EF % (78.9 ± 1.4 % vs. 81.72 ± 1.72 % pre-I/R), but this was not statistically significant ($P = 0.97$ vs pre-I/R). This is likely due to residual effects of the surgical intervention, anaesthesia, and analgesia. At D28 post-I/R (**Fig. 7C**) the sham-operated mice had returned to normal pre-IR EF % levels of 81.0 ± 0.6 %, as expected. The mice which received vehicle control injections had an average EF% of 44.3 ± 3.2 %, whereas mice which received SCPL-EVs had an average EF% of 63.4 ± 4.9 %. This difference was statistically significant ($P = 0.0048$, one-way ANOVA). Calculating the change in EF % of each individual mouse from D1 to D28 (**Fig. 7D**) showed that vehicle control-treated mice showed average deterioration of -8.2 ± 2.9 %. Notably, none of the vehicle-treated mice showed any improvement in EF % from D1 to D28. This reduction in cardiac function is expected following I/R injury due to cardiomyocyte loss and fibrotic tissue formation. On the other hand, mice treated with SCPL-EVs showed an average improvement in EF % of $+11.9 \pm 2.7$ % from D1 to D28. This was statistically significant when compared against vehicle control-treated animals ($P = 0.0001$, one-way ANOVA), indicating a strong therapeutic effect of the SCPL-EVs. Each group was also compared against an expected change of zero using one-sample t-tests. This showed that the reduction in EF % in the I/R + vehicle group and the increase in EF % in the I/R + SCPL-EV group over 28 days were both statistically significant. Additional echocardiography metrics (**Supplemental Fig. 7**) show that I/R mice showed a larger left ventricle internal diameter at diastole (LVIDd) and systole (LVIDs) than the sham mice. This is due to dilation of the ventricle which occurs following loss of myocardium in the ventricle wall. For both measurements, SCPL-EV-treated animals showed a smaller LVID ($P = 0.0123$ and $P = 0.0138$) than vehicle control-treated animals at D28, demonstrating that SCPL-EVs were able to reduce the degree of ventricular dilation. This was also supported by the pressure-volume loops (**Supplemental Fig. 7E**) which showed less ventricular dilation and greater pressure generation in SCPL-EV-treated animals.

There was no significant change in mouse heart weight as a percentage of body weight (**Fig. 7E**), nor any change in overall mouse body weight (**Fig. 7F**). There were also no differences in organ weights which would indicate toxicity or a severe immune response (**Fig. 7G**). There was a slight reduction in liver weight (-10.7 % vs. sham, $p = 0.002$) in the I/R + vehicle group, which was not present in the I/R + SCPL-EV group ($p = 0.99$ vs. sham). Intraventricular catheter was used to directly measure contractile function of the left ventricle prior to sacrifice. dP/dT maximum and minimum values are shown in **Fig. 7H**. The results showed the same trend as echocardiography, with I/R + vehicle animals demonstrating poorer cardiac function than sham-operated mice. Mice which received I/R + SCPL-EV showed significantly improved dP/dT values compared to vehicle control-treated animals ($P = 0.0072$ for max, $P = 0.0432$ for min). Lastly, cross sections of the heart were stained with Masson's trichrome to reveal the amount of collagen scar formation. The images (**Fig. 7I**) showed that mice in the I/R + Vehicle group had thin left ventricle walls, which were composed of mainly fibrous tissue, which is expected following I/R and remodelling. The left ventricle wall thickness of sham mice was 842 ± 44.1 μ m compared to 332.7 ± 86.39 μ m for vehicle-treated I/R mice. Mice which received SCPL-EVs had preserved wall thickness (858.5 ± 39.23 , $P = 0.015$, nested one-way ANOVA) (**Fig. 7J**). Scar length was also greatly reduced from 4547 ± 212.9 μ m to 2126 ± 68.88 μ m ($P = 0.0002$, one-way ANOVA with Tukey's post-test). This preservation of cardiac muscle tissue is in line with the improved cardiac function of the SCPL-EV-treated mice (**Fig. 7C–H**).

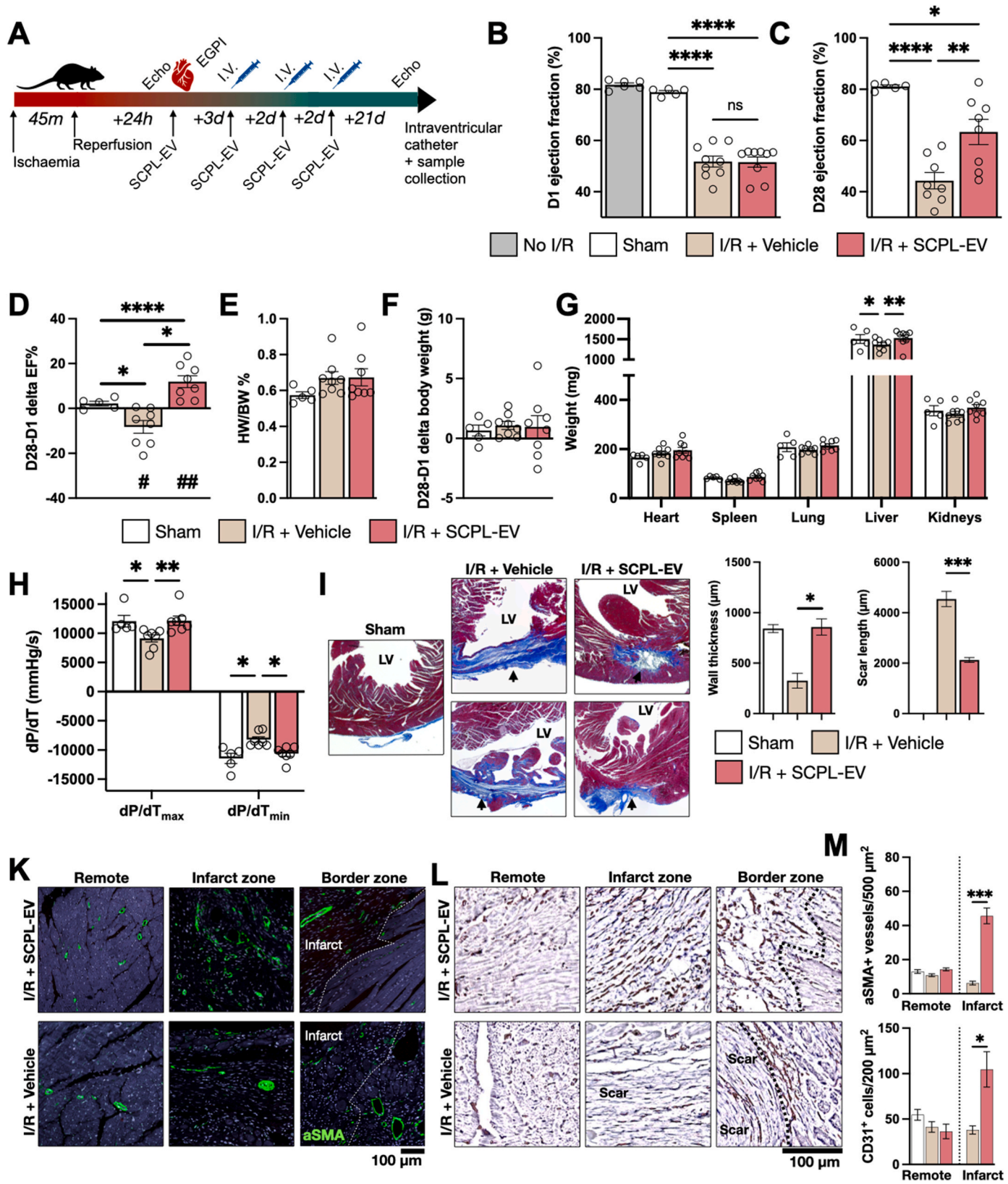
Lastly, we carried out staining for alpha-SMA (myofibroblasts and smooth muscle) (**Fig. 7K**) and CD31 (endothelial cells) (**Fig. 7L**) and examined the remote area, infarct zone and border zone areas. The results (**Fig. 7M**) showed a significantly higher density of a-SMA-positive vessels and CD31-positive endothelial cells in the infarct zone following SCPL-EV treatment. In the vehicle-control treated animals, the infarct zone comprised mostly a thin band of collagenous scar tissue with very few a-SMA⁺ vessels or CD31⁺ cells. Together, the results of this experiment clearly showed that SCPL-EVs were able to preserve, and even improve, cardiac function following I/R injury. This likely occurs through multiple mechanisms including protection of cardiomyocytes, stimulation of angiogenesis, and immunomodulation. Furthermore, the SCPL-EVs appeared to be well tolerated, even in a xenobiotic model.

4. Discussion

In summary, this study has shown that human clinical-grade PCs can be used to produce EVs which can emerge as a therapeutic biomaterial for myocardial infarction therapy. SCPL-EVs had strong protective activities in *in vitro* hypoxia/re-oxygenation injury assays, preserving cardiomyocyte metabolic activity and reducing apoptosis with equal potency to BM-MSC EVs. SCPL-EVs also stimulated cardiac microvessel endothelial cell tube formation, whereas the pre-fractionation SCPL-W did not, likely due to the differences in protein and miRNA composition between the two materials. In mice, SCPL-EVs robustly improved cardiac function, reduced scar formation and promoted angiogenesis following I/R injury, with no obvious adverse effects.

There are several major advantages to using PCs as a starting product for therapeutic EV generation. Firstly, in our study, the PCs were obtained from a national blood establishment; thus, the starting product was a licensed, clinical-grade product suitable for human transfusion. These were obtained using standard collection and storage procedures, including donor screening and viral testing, ensuring consistent platelet counts, quality, purity and safety. Secondly, while platelet properties and their EVs likely do vary between individuals, PCs or PC-derived fractions are easily pooled to include donations from multiple healthy subjects, minimising the effects of individual donor variation. Our study generated SCPL-EVs from 15 to 18 individual healthy donors, a number sufficient to ensure a constant concentration of growth factors [37]. Lastly, PCs are available worldwide, making SCPL-EVs an accessible source of therapeutic EVs. mRNA and DNA concentrations were low in SCPL and were lowered further in SCPL-EV fractions. Additionally, these nucleic acids have not presented any safety issues in use of allogeneic platelet transfusions [29]. Together, these factors should allow for faster regulatory approval globally. The platelet activation and EV isolation procedures allowed for sterile handling procedures and a high degree of standardisation, and we also found that size-exclusion chromatography produced highly repeatable and consistent results across multiple batches, with very little inter-operator variability; in agreement with a previous study [39].

Our *in vitro* assays confirmed that the SCPL-EVs had pro-angiogenic and cardiomyocyte protective activities on human and rodent cardiac cells and offered efficacy similar to BM-MSC EVs. However, given that SCPL-EVs are contained within SCPL-W, we were surprised that SCPL-W was much less effective at stimulating angiogenesis or protecting H/R cardiomyocytes. Our results indicate that this is likely due to the different protein and miRNA cargo of the SCPL-EV fraction. Proteomics showed that the EV fraction had reduced coagulation factors, carrier proteins and complement proteins, which may have inhibitory effects on angiogenesis. For example, platelets are known to contain thrombospondin 1, angiostatin, plasminogen activation inhibitor 1 and platelet factor 4 (CXCL4); all of which are angiogenesis inhibitors [69]. Our proteomic data should not be interpreted as a complete list of all proteins contained within SCPL-EVs since there are some limits in the sensitivity of this technique compared to protein targeted assays [30, 38]. An antibody-based multiplex array detected IGF-1, epidermal



(caption on next page)

Fig. 7. Efficacy of SCPL-EVs as a therapeutic following cardiac ischaemia/reperfusion (I/R) injury

- A. Schematic diagram of cardiac ischaemia/reperfusion (I/R) injury followed by SCPL-EV administration by echo-guided percutaneous intracavitary (EGPI) administration at 24h, and three subsequent intravenous injections at D3, D5 and D7.
- B. Ejection fraction % (EF%) of mice 24h following sham I/R or I/R surgery, prior to commencement of treatment. N = 6 mice without I/R are shown for reference of normal EF%. Sham (n = 5), I/R + vehicle (n = 9) and I/R + SCPL-EV (n = 9) groups were compared by one-way ANOVA with Tukey's post-test.
- C. EF% of mice 28 days following cardiac I/R injury. Groups were compared by one-way ANOVA with Tukey's post-test.
- D. Change in EF% (Δ EF%) for individual mice from D1 to D28. Groups were compared by one-way ANOVA with Tukey's post-test, indicated by asterisks above the bars. Each group was also compared by one-sample t-test against an expected change of zero, indicated by # below the bars.
- E. Heart weight (HW) as a percentage of body weight (BW).
- F. Change in mouse body weight from D1 to D28 post-surgery.
- G. Organ weights of animals from sham, I/R + vehicle and I/R + SCPL-EV groups. Groups were compared by two-way ANOVA.
- H. dP/dT max/min calculated from pressure-volume readings by intraventricular catheter. Groups were compared by two-way ANOVA with Sidak's multiple comparison test.
- I. Representative Masson's Trichrome-stained cross sections of hearts from sham, I/R + Vehicle and I/R + SCPL-EV mice. The left ventricle wall is indicated by an arrow and the chamber is annotated with LV.
- J. Quantification of wall thickness and scar length based on Trichrome-stained images. Wall thickness was compared using nested ANOVA to correct for multiple measurements per sample. Vehicle and SCPL-EV group scar lengths were compared by one-way ANOVA.
- K. Representative immunofluorescence staining for alpha-SMA (green) in the remote, infarct and border zones of SCPL-EV or vehicle control-treated mice. A light microscope image is overlaid for visualisation of the tissue structure. In the border zone image, the margin of infarct area is annotated. Scale bar 100 μ m.
- L. Representative immunohistochemical staining for CD31 (brown), counterstained with haematoxylin in the remote, infarct and border zones of SCPL-EV or vehicle control-treated mice. Scar tissue in the vehicle group is annotated. Scale bar 100 μ m.
- M. Quantification of alpha-SMA⁺ vessels (with visible lumen), and CD31⁺ cell density in the remote and infarct regions. * = $P \leq 0.05$, ** = $P \leq 0.01$ and *** = $P \leq 0.001$. ns = not significant ($P > 0.05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

growth factor (EGF), platelet-derived growth factor (PDGF), interleukins and several matrix metalloproteases (MMPs) in SCPL-EVs. This agrees with previous studies [30,38].

In terms of *in vivo* efficacy, our outcomes are very similar to a recently reported study of mouse iPSC-derived EVs using the same mouse strain, I/R model and similar EV dose as our study [70]. Adamiak and colleagues' I/R model was milder (30 min occlusion compared to 45 min in our study), but the proportional increase in EF% after 28 days was very similar between our two studies. Our results also are also comparable to those using human iPSC EVs and human cardiosphere cell-derived EVs [62,71]. This indicates that SCPL-EVs compare favourably to other known therapeutic EV types.

During our investigation we noted the poor overlap between the SCPL-EV proteomic profile and public proteomic databases of EVs derived from cultured cells and body fluids. As described earlier (Supplemental Fig. 3), this is likely due to lower numbers of P-EV studies included in these databases. However, this also illustrates that P-EVs are biologically distinct from many other circulating or cell-secreted EVs, and thus carry their own specific cargo. The same was true for the miRNA profile. We found that SCPL-EVs had more overlap with serum EVs (58%) than BM MSC-EVs (46%); this should be expected, since some serum EVs are derived from platelets. However, we also found that SCPL-EVs also contained a high proportion of miRNAs (38%) which were not detected in serum or stem cell-secreted EVs, again indicating that they are a biologically distinct population. The cargo may also be influenced by platelet activation, storage, and processing methods [72]. Our analysis (Supplemental Table 4) indicated that many of the highest abundance SCPL miRNAs have been shown to play anti-inflammatory, immunomodulatory roles. Our *in vitro* analysis of macrophage polarisation following SCPL-W and SCPL-EV treatment also supported this, with both preparations inducing a M2-type phenotype and significantly increasing *Il10* expression. It is also notable that SCPL-EVs contained several well-known cardioprotective miRNAs including miR-223, miR-142, miR-21, miR-19a, and miR-23a-3p [11,60,73–76]. However, while single miRNAs can have potent protective activities, one advantage of EVs is their ability to deliver complex cargos which act upon multiple targets. As such, it is the combined interacting effects of potentially hundreds of EV cargo components which likely bring about multiple biological changes. Precise roles of SCPL miRNAs could be investigated in future studies.

Although our study demonstrated strong therapeutic benefits in a mouse model, there are some limitations which should be considered. Firstly, we administered human EVs in mice which may induce a

xenobiotic immune response, particularly after multiple injections. This compromise was necessary because immunocompromised mice have poor survival and inconsistent cardiac function following MI, and immunosuppression interferes with MI recovery processes [77]. Therefore we elected to use C57BL/6 mice, since they are the preferred mouse strain in the cardiovascular field. This was also part of our rationale in testing purified SCPL-EVs which had less complement proteins and coagulation factors than SCPL-W. We did not observe any notable adverse changes in terms of body weight, organ weight, histological appearance, or blood coagulation parameters in our experiment animals; however, it is likely that some xeno-immunisation would still occur. Despite this, SCPL-EVs produced a robust and statistically significant improvement in cardiac function 28 days after cardiac I/R injury. We also note that trials for regulatory approval would ask for using human EVs in immunocompetent animal models. PCs, containing EVs, are frequently given intravenously to human patients, and a recent publication presented data from a human phase I clinical trial showing that P-EVs, isolated from allogeneic PCs screened for ABO and RhD groups, red blood cell antibody, and transfusion-transmissible viruses, as routinely done, were well-tolerated when applied to a topical injury site [33]. As such, we do not anticipate that there would be any major safety concerns administering allogeneic SCPL-EVs in human patients. If needed, PCs can be subjected to additional, already-licensed pathogen reduction treatments using photochemical inactivation [78].

In our study we noted a relatively high concentration of SCPL-EVs remaining in the blood 30 min after IV injection compared to other studies [79,80]. We speculate that this may be due to SCPL-EVs originating from platelets, thus expressing protective surface markers such as CD47. Supporting this, several studies have shown that platelet membrane-coating can extend circulation time of nanocarriers and improve homing to the infarcted myocardium. Platelet-coated nanoparticles can also home to injury sites by binding to circulating monocytes [81,82]. Exploring the pharmacokinetics of SCPL-EVs and more precisely measuring biodistribution at different timepoints will be a subject of our future studies. In this study, delivery of SCPL-EVs to the infarcted myocardium was approximately 0.8 % after a single injection, which is within the range of other studies of EVs and nanomedicines [64]. However, this concentration still leaves considerable room for improvement; perhaps by combining the EVs with a hydrogel, nanofiber patch, decellularised matrix, or other material which can provide controlled release [83,84]. Cell-specific uptake depends on the EV origins, since this determines EV surface proteins and binding efficiency to different complementary cell membrane proteins. In our study we

confirmed rapid uptake of SCPL-EVs by human cardiomyocytes in culture, but we did not compare with other cardiac cell types. A previous study interestingly showed that cardiosphere-derived cell EVs were taken up mainly by cardiac fibroblasts and immune cells rather than cardiomyocytes following MI [66]. We believe it is likely that multiple cell types in the heart take up the SCPL-EVs after I/R, including cardiac fibroblasts, endothelial cells, cardiomyocytes and immune cells such as macrophages [63]. It is also possible that EVs can be taken up by circulating cells which may also alter their function or deliver them to the infarct area. Since platelets are known to interact with circulating monocytes following myocardial infarction, determining whether P-EVs have the same ability could be a subject for future research.

5. Conclusion

This study has demonstrated that EVs derived from minimal processing of human clinical-grade platelet lysates have powerful therapeutic properties following myocardial ischaemia/reperfusion injury. Profiled proteomic and miRNA cargo of these EVs revealed that they share similarities with other EV populations, but also carry their own specific cargo. The EV production and separation process described in this manuscript is scalable, and platelet-derived biomaterials have an excellent safety profile and established infrastructure surrounding their use. As such, therapies using platelet-derived EVs could be rapidly translated into human clinical trials.

Funding sources

Work in this manuscript was supported by Taiwan National Science & Technology Council: NSTC 112-2636-B-038-005 (DJL) and 112-2923-E-038-001 (TB). This work was financially supported of the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan: DP2-TMU-112-O-02 (DJL and TB).

CRediT authorship contribution statement

Dora Livkisa: Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Validation. **Tzu-Hsin Chang:** Data curation, Investigation. **Thierry Burnouf:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – original draft, Writing – review & editing. **Andreas Czossek:** Investigation. **Nhi Thao Ngoc Le:** Investigation, Visualization. **Gleb Shamrin:** Investigation. **Wei-Ting Yeh:** Investigation. **Masao Kamimura:** Supervision, Writing – review & editing. **David J. Lundy:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Thierry Burnouf is a co-founder of a company (Invenis Therapeutics) developing platelet lysates for neurological disorders, and is a co-inventor of patents relating to therapeutic platelet lysates, owned by Taipei Medical University, Taiwan, and the University of Lille, France. The other authors have no conflicts of interest to declare.

Data availability

Data will be made available on request.

Acknowledgements

The authors thank the staff at Preclinical Core Facility for Cardiovascular and Stroke Animal Models, Academia Sinica Institute of Biomedical Science, Taiwan, for their technical support in I/R surgeries,

echocardiography, and intraventricular catheter measurements. The authors thank the Taipei Blood Center (Guandu, Taiwan Blood Services Foundation, Taiwan) for the supply of PCs. We are grateful to the Instrumentation Center and Consortia of Key Technologies of National Taiwan University for their technical research assistance in MS. The authors thank Ms. Ariunjargal Nyam-Erdene for providing the SCPL, and Mr. Manuel SV Jaimes for his preliminary work on the study. The authors thank Wei-Ting Huang at TMU Core Facility for assistance with IVIS. The cryo-EM experiments were performed at the Academia Sinica Cryo-EM Facility (ASCEM). ASCEM is jointly supported by Academia Sinica Core Facility and Innovative Instrument Project (Grant No. AS-CFII-111-210) and Taiwan Protein Project (Grant No. AS-KPQ-109-TPP2). We also thank Dr. Chia-Te Liao, Shuang-Ho Hospital, Taipei, for supporting BMDM isolation and Ms. Helen Nguyen for assistance with BM-MSC EV ultracentrifugation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biomaterials.2024.122502>.

References

- [1] P.K. Mishra, A. Adameova, J.A. Hill, C.P. Baines, P.M. Kang, J.M. Downey, J. Narula, M. Takahashi, A. Abbate, H.C. Piristine, S. Kar, S. Su, J.K. Higa, N. K. Kawasaki, T. Matsui, Guidelines for evaluating myocardial cell death, *Am. J. Physiol. Heart Circ. Physiol.* 317 (2019) 891–922, <https://doi.org/10.1152/ajpheart.00259.2019>.
- [2] A. Prasad, G.W. Stone, D.R. Holmes, B. Gersh, Reperfusion injury, microvascular dysfunction, and cardioprotection: the “dark side” of reperfusion, *Circulation* 120 (2009) 2105–2112, <https://doi.org/10.1161/CIRCULATIONAHA.108.814640>.
- [3] O. Bergmann, S. Zdunek, A. Felker, M. Salehpour, K. Alkass, S. Bernard, S. L. Sjöström, M. Szewczykowska, T. Jackowska, C. dos Remedios, T. Malm, M. Andrä, R. Jashari, J.R. Nyengaard, G. Possnert, S. Jovinge, H. Druid, J. Frisén, Dynamics of cell generation and turnover in the human heart, *Cell* (2015) 1566–1575, <https://doi.org/10.1016/j.cell.2015.05.026>.
- [4] J. Li, Y. Zhao, W. Zhu, Targeting angiogenesis in myocardial infarction: novel therapeutics, *Exp. Ther. Med.* 23 (2021), <https://doi.org/10.3892/etm.2021.10986>.
- [5] Z. Tao, B. Chen, X. Tan, Y. Zhao, L. Wang, T. Zhu, K. Cao, Z. Yang, Y.W. Kan, H. Su, Coexpression of VEGF and angiotensin-1 promotes angiogenesis and cardiomyocyte proliferation reduces apoptosis in porcine myocardial infarction (MI) heart, *Proc. Natl. Acad. Sci. U. S. A.* 108 (2011) 2064–2069, <https://doi.org/10.1073/pnas.1018925108>.
- [6] M.A. Borrelli, H.R. Turnquist, S.R. Little, Biologics and their delivery systems: trends in myocardial infarction, *Adv. Drug Deliv. Rev.* 173 (2021) 181–215, <https://doi.org/10.1016/j.addr.2021.03.014>.
- [7] S. Huang, N.G. Frangogiannis, Anti-inflammatory therapies in myocardial infarction: failures, hopes and challenges, *Br. J. Pharmacol.* 175 (2018) 1377–1400, <https://doi.org/10.1111/bph.14155>.
- [8] K. Malliaras, R.R. Makkar, R.R. Smith, K. Cheng, E. Wu, R.O. Bonow, L. Marbán, A. Mendizabal, E. Cingolani, P.V. Johnston, G. Gerstenblith, K.H. Schuleri, A. C. Lardo, E. Marbán, Intracoronary cardiosphere-derived cells after myocardial infarction, *J. Am. Coll. Cardiol.* 63 (2014) 110–122, <https://doi.org/10.1016/j.jacc.2013.08.724>.
- [9] T.-T. Cung, O. Morel, G. Cayla, G. Rioufol, D. Garcia-Dorado, D. Angoulvant, E. Bonnefoy-Cudraz, P. Guérin, M. Elbaz, N. Delarche, P. Coste, G. Vanzetto, M. Metge, J.-F. Aupetit, B. Jouve, P. Motreff, C. Tron, J.-N. Labeque, P.G. Steg, Y. Cottin, G. Range, J. Clerc, M.J. Claeys, P. Coussement, F. Prunier, F. Moulin, O. Roth, L. Belle, P. Dubois, P. Barragan, M. Gilard, C. Piot, P. Colin, F. De Poli, M.-C. Morice, O. Ider, J.-L. Dubois-Randé, T. Untersee, H. Le Breton, T. Béard, D. Blanchard, G. Grollier, V. Malquart, P. Staat, A. Sudre, E. Elmer, M.J. Hansson, C. Bergerot, I. Boussaha, C. Jossan, G. Derumeaux, N. Mewton, M. Ovize, Cyclosporine before PCI in patients with acute myocardial infarction, *N. Engl. J. Med.* 373 (2015) 1021–1031, <https://doi.org/10.1056/nejmoa1505489>.
- [10] V. Schächinger, S. Erbs, A. Elsässer, W. Haberbusch, R. Hambrecht, H. Hölschermann, J. Yu, R. Corti, D.G. Mathey, C.W. Hamm, T. Süselbeck, B. Assmus, T. Tonn, S. Dimmeler, A.M. Zeiher, Intracoronary bone marrow-derived progenitor cells in acute myocardial infarction, *N. Engl. J. Med.* 355 (2006) 1210–1221, <https://doi.org/10.1056/nejmoa060186>.
- [11] G.-L. Gu, X.-L. Xu, X.-T. Sun, J. Zhang, C.-F. Guo, C.-S. Wang, B. Sun, G.-L. Guo, K. Ma, Y.-Y. Huang, L.-Q. Sun, Y.-Q. Wang, Cardioprotective effect of MicroRNA-21 in murine myocardial infarction, *Cardiovasc. Ther.* 33 (2015) 109–117, <https://doi.org/10.1111/1755-5922.12118>.
- [12] M. Tkach, C. Théry, Communication by extracellular vesicles: where we are and where we need to go, *Cell* 164 (2016) 1226–1232, <https://doi.org/10.1016/j.cell.2016.01.043>.
- [13] J. Kowal, G. Arras, M. Colombo, M. Jouve, J.P. Morath, B. Primdal-Bengtson, F. Dingli, D. Loew, M. Tkach, C. Théry, Proteomic comparison defines novel

- markers to characterize heterogeneous populations of extracellular vesicle subtypes, *Proc. Natl. Acad. Sci. USA* 113 (2016), <https://doi.org/10.1073/pnas.1521230113>.
- [14] L. Barile, T. Moccetti, E. Marb n, G. Vassalli, Roles of exosomes in cardioprotection, *Eur. Heart J.* 38 (2017) 1372–1379, <https://doi.org/10.1093/eurheartj/ehw304>.
- [15] R. Gallet, J. Dawkins, J. Valle, E. Sinsolo, G. De Couto, R. Middleton, E. Tseliou, D. Luthringer, M. Kreke, R.R. Smith, L. Marb n, B. Ghaleh, E. Marb n, Exosomes secreted by cardiosphere-derived cells reduce scarring, attenuate adverse remodelling, and improve function in acute and chronic porcine myocardial infarction, *Eur. Heart J.* 38 (2017) 201–211, <https://doi.org/10.1093/eurheartj/ehw240>.
- [16] S.M. Davidson, C.M. Boulanger, E. Aikawa, L. Badimon, L. Barile, C.J. Binder, A. Brisson, E. Buzas, C. Emanueli, F. Jansen, M. Katsur, R. Lacroix, S.K. Lim, N. Mackman, M. Mayr, P. Menasch , R. Nieuwland, S. Sahoo, K. Takov, T. Thum, P. Vader, M.H.M. Wauben, K. Witwer, J.P.G. Sluijter, Methods for the identification and characterization of extracellular vesicles in cardiovascular studies: from exosomes to microvesicles, *Cardiovasc. Res.* (2022), <https://doi.org/10.1093/cvr/cvac031>.
- [17] K.W. Witwer, E.I. Buz s, L.T. Bemis, A. Bora, C. L sser, J. L tvall, E.N. Nolte-‐t Hoen, M.G. Piper, S. Sivaraman, J. Skog, C. Th ry, M.H. Wauben, F. Hochberg, Standardization of sample collection, isolation and analysis methods in extracellular vesicle research, *J. Extracell. Vesicles* 2 (2013), <https://doi.org/10.3402/jev.v2i0.20360>.
- [18] A. Czosseck, M.M. Chen, H. Nguyen, A. Meeson, C. Hsu, C. Chen, T.A. George, S. Ruan, Y. Cheng, P. Lin, P.C.H. Hsieh, D.J. Lundy, Porous scaffold for mesenchymal cell encapsulation and exosome-based therapy of ischemic diseases, *J. Contr. Release* 352 (2022) 879–892, <https://doi.org/10.1016/j.jconrel.2022.10.057>.
- [19] T. Burnouf, V. Aghahari, V. Aghahari, Extracellular vesicles as nanomedicine: hopes and hurdles in clinical translation, *Int. J. Nanomed.* 14 (2019) 8847–8859, <https://doi.org/10.2147/IJN.S225453>.
- [20] H. Wang, R. Maimaitiaili, J. Yao, Y. Xie, S. Qiang, F. Hu, X. Li, C. Shi, P. Jia, H. Yang, M. Wei, J. Zhao, Z. Zhou, J. Xie, J. Jiang, H. Cai, J.P.G. Sluijter, Y. Xu, Y. Zhang, J. Xiao, Percutaneous intracoronary delivery of plasma extracellular vesicles protects the myocardium against ischemia-reperfusion injury in Canis, *Hypertension* (2021) 1541–1554, <https://doi.org/10.1161/HYPERTENSIONAHA.121.17574>.
- [21] J.M. Vicencio, D.M. Yellon, V. Sivaraman, D. Das, C. Boi-Doku, S. Arjun, Y. Zheng, J.A. Riquelme, J. Kearney, V. Sharma, G. Multhoff, A.R. Hall, S.M. Davidson, Plasma exosomes protect the myocardium from ischemia-reperfusion injury, *J. Am. Coll. Cardiol.* 65 (2015) 1525–1536, <https://doi.org/10.1016/j.jacc.2015.02.026>.
- [22] M.S.V. Jaimes, C. Liao, M.M. Chen, A. Czosseck, T. Lee, Y. Chou, Y. Chen, S. Lin, J. Lai, D.J. Lundy, Assessment of circulating extracellular vesicles from calorie-restricted mice and humans in ischaemic injury models, *J. Extracell. Biol.* 2 (2023), <https://doi.org/10.1002/jex2.86>.
- [23] H. Zhao, X. Chen, G. Hu, C. Li, L. Guo, L. Zhang, F. Sun, Y. Xia, W. Yan, Z. Cui, Y. Guo, X. Guo, C. Huang, M. Fan, S. Wang, F. Zhang, L. Tao, Small extracellular vesicles from Brown adipose tissue mediate exercise cardioprotection, *Circ. Res.* 130 (2022) 1490–1506, <https://doi.org/10.1161/CIRCRESAHA.121.320458>.
- [24] H. Zhang, S. Lin, C.L. McElroy, B. Wang, D. Jin, V.V. Uteshev, K. Jin, Circulating pro-inflammatory exosomes worsen Stroke outcomes in aging, *Circ. Res.* 129 (2021) 121–140, <https://doi.org/10.1161/CIRCRESAHA.121.318897>.
- [25] M.S. Mahts, M.C. K ller, K. Papp, C. Aigner, D. Dioso, P. Mucher, H. Schachner, M. Bonelli, H. Haslacher, A.J. Rees, R. Kain, Small extracellular vesicles are released ex vivo from platelets into serum and from residual blood cells into stored plasma, *J. Extracell. Biol.* 2 (2023), <https://doi.org/10.1002/jex2.88>.
- [26] J.W.P. Bracht, M. Los, M.A.J. van Eijndhoven, B. Bettin, E. van der Pol, D. M. Pegtel, R. Nieuwland, Platelet removal from human blood plasma improves detection of extracellular vesicle-associated miRNA, *J. Extracell. Vesicles* 12 (2023), <https://doi.org/10.1002/jev2.12302>.
- [27] R.J. Berckmans, R. Lacroix, C.M. Hau, A. Sturk, R. Nieuwland, Extracellular vesicles and coagulation in blood from healthy humans revisited, *J. Extracell. Vesicles* 8 (2019), <https://doi.org/10.1080/20013078.2019.1688936>.
- [28] J. Johnson, Y.W. Wu, C. Blyth, G. Lichtfuss, H. Goubran, T. Burnouf, Prospective therapeutic applications of platelet extracellular vesicles, *Trends Biotechnol.* 39 (2021) 598–612, <https://doi.org/10.1016/j.tibtech.2020.10.004>.
- [29] K. Schallmoser, R. Henschler, C. Gabriel, M.B.C. Koh, T. Burnouf, Production and quality requirements of human platelet lysate: a position statement from the working party on cellular therapies of the International Society of blood transfusion, *Trends Biotechnol.* 38 (2020) 13–23, <https://doi.org/10.1016/j.tibtech.2019.06.002>.
- [30] T. Burnouf, M.-L. Chou, D.J. Lundy, E.-Y. Chuang, C.-L. Tseng, H. Goubran, Expanding applications of allogeneic platelets, platelet lysates, and platelet extracellular vesicles in cell therapy, regenerative medicine, and targeted drug delivery, *J. Biomed. Sci.* 30 (2023) 79, <https://doi.org/10.1186/s12929-023-00972-w>.
- [31] V. Aghahari, V. Aghahari, P.A. Burnouf, C.H. Chew, T. Burnouf, Extracellular microvesicles as new industrial therapeutic frontiers, *Trends Biotechnol.* 37 (2019) 707–729, <https://doi.org/10.1016/j.tibtech.2018.11.012>.
- [32] H.D. Flad, E. Brandt, Platelet-derived chemokines: pathophysiology and therapeutic aspects, *Cell. Mol. Life Sci.* 67 (2010) 2363–2386, <https://doi.org/10.1007/s00018-010-0306-x>.
- [33] J. Johnson, S.Q.K. Law, M. Shojaaee, A.S. Hall, S. Bhuiyan, M.B.L. Lim, A. Silva, K.J. W. Kong, M. Schoppet, C. Blyth, H.N. Ranasinghe, N. Sejjic, M.J. Chuei, O. C. Tatford, A. Cifuentes-Rius, P.F. James, A. Tester, I. Dixon, G. Lichtfuss, First-in human clinical trial of allogeneic, platelet-derived extracellular vesicles as a potential therapeutic for delayed wound healing, *J. Extracell. Vesicles* 12 (2023), <https://doi.org/10.1002/jev2.12332>.
- [34] K. Stellos, B. Bigalke, O. Borst, F. Pfaff, A. Elskamp, S. Sachsenmaier, R. Zachmann, K. Stamatiopoulos, T. Sch nberger, T. Geisler, H. Langer, M. Gawaz, Circulating platelet-progenitor cell coaggregate formation is increased in patients with acute coronary syndromes and augments recruitment of cd341 cells in the ischaemic microcirculation, *Eur. Heart J.* 34 (2013) 2548–2556, <https://doi.org/10.1093/eurheartj/ehf131>.
- [35] J. Tang, T. Su, K. Huang, P.U. Dinh, Z. Wang, A. Vandergriff, M.T. Hensley, J. Cores, T. Allen, T. Li, E. Sproul, E. Mihalko, L.J. Lobo, L. Ruterbories, A. Lynch, A. Brown, T.G. Caranasos, D. Shen, G.A. Stouffer, Z. Gu, J. Zhang, K. Cheng, Targeted repair of heart injury by stem cells fused with platelet nanovesicles, *Nat. Biomed. Eng.* 2 (2018) 17–26, <https://doi.org/10.1038/s41551-017-0182-x>.
- [36] B. Cheng, E.K.W. Toh, K.-H. Chen, Y.-C. Chang, C.-M.J. Hu, H.-C. Wu, L.-Y. Chau, P. Chen, P.C.H. Hsieh, Biomimicking platelet-monocyte interactions as a novel targeting strategy for heart healing, *Adv. Healthcare Mater.* (2016), <https://doi.org/10.1002/adhm.201600724>.
- [37] A. Bianchetti, C. Chinello, M. Guindani, S. Braga, A. Neva, R. Verardi, G. Piovani, L. Pagani, G. Lisignoli, F. Magni, D. Russo, C. Almici, A blood bank standardized production of human platelet lysate for mesenchymal stromal cell expansion: proteomic characterization and biological effects, *Front. Cell Dev. Biol.* 9 (2021), <https://doi.org/10.3389/fcell.2021.650490>.
- [38] A. Nyam-Erdene, O. Nebie, L. Delila, L. Bu e, D. Devos, S.Y. Chou, D. Blum, T. Burnouf, Characterization and chromatographic isolation of platelet extracellular vesicles from human platelet lysates for applications in neuroregenerative medicine, *ACS Biomater. Sci. Eng.* 7 (2021) 5823–5835, <https://doi.org/10.1021/acsbiomaterials.1c01226>.
- [39] P.M. Vanderboom, S. Dasari, G.N. Rueggsegger, M.W. Pataky, F. Lucien, C. J. Heppelmann, I.R. Lanza, K.S. Nair, A size-exclusion-based approach for purifying extracellular vesicles from human plasma, *Cell Rep. Meth.* 1 (2021) 100055, <https://doi.org/10.1016/j.crmeth.2021.100055>.
- [40] B. Benayas, J. Morales, C. Egea, P. Armi sen, M. Y  ez-M  , Optimization of extracellular vesicle isolation and their separation from lipoproteins by size exclusion chromatography, *J. Extracell. Biol.* 2 (2023), <https://doi.org/10.1002/jex2.100>.
- [41] Y. Guo, A. Tomlin, X. Zhu, M. Wysoczynski, R. Bolli, Echocardiography-guided percutaneous left ventricular intracavitary injection as a cell delivery approach in infarcted mice, *Mol. Cell Biochem.* (2021) 476, <https://doi.org/10.1007/s11010-021-04077-6>.
- [42] R. Widyaniangrum, Y.-W. Wu, L. Delila, D.-Y. Lee, T.-J. Wang, T. Burnouf, In vitro evaluation of platelet extracellular vesicles (PEVs) for corneal endothelial regeneration, *Platelets* 33 (2022) 1237–1250, <https://doi.org/10.1080/09537104.2022.2105829>.
- [43] G. Toda, T. Yamauchi, T. Kadowaki, K. Ueki, Preparation and culture of bone marrow-derived macrophages from mice for functional analysis, *STAR Prot.* 2 (2021) 100246, <https://doi.org/10.1016/j.xpro.2020.100246>.
- [44] A. Tojo, S. Kinugasa, Mechanisms of glomerular albumin filtration and tubular reabsorption, *Internat. J. Nephrol.* 2012 (2012) 481–520, <https://doi.org/10.1155/2012/481520>.
- [45] J. Webber, A. Clayton, How pure are your vesicles? *J. Extracell. Vesicles* 2 (2013) 1–6, <https://doi.org/10.3402/jev.v2i0.19861>.
- [46] J. Moss, R. Ben-Ami, E. Shai, O. Gal-Rosenberg, Y. Kalish, A. Klochendler, G. Cann, B. Glaser, A. Arad, R. Shemer, Y. Dor, Megakaryocyte- and erythroblast-specific cell-free DNA patterns in plasma and platelets reflect thrombopoiesis and erythropoiesis levels, *Nat. Commun.* 14 (2023), <https://doi.org/10.1038/s41467-023-43310-2>.
- [47] M.R. Fernando, C. Jiang, G.D. Krzyzanowski, W.L. Ryan, New evidence that a large proportion of human blood plasma cell-free DNA is localized in exosomes, *PLoS One* 12 (2017), <https://doi.org/10.1371/journal.pone.0183915>.
- [48] H.J. Kim, M.J. Rames, F. Goncalves, C.W. Kirschbaum, B. Roskams-Hieter, E. Spiliotopoulos, J. Briand, A. Doe, J. Estabrook, J.T. Wagner, E. Demir, G. Mills, T.T.M. Ngo, Selective enrichment of plasma cell-free messenger RNA in cancer-associated extracellular vesicles, *Commun. Biol.* 6 (2023), <https://doi.org/10.1038/s42003-023-05232-z>.
- [49] Y. Yuana, R.I. Koning, M.E. Kuil, P.C.N. Rensen, A.J. Koster, R.M. Bertina, S. Osanto, Cryo-electron microscopy of extracellular vesicles in fresh plasma, *J. Extracell. Vesicles* 2 (2013) 21494, <https://doi.org/10.3402/jev.v2i0.21494>.
- [50] F. Bonavita, C. Stefanelli, E. Giordano, M. Columbaro, A. Facchini, F. Bonaf , C. M. Caldara, C. Guarnieri, H9c2 cardiac myoblasts undergo apoptosis in a model of ischemia consisting of serum deprivation and hypoxia: inhibition by PMA, *FEBS Lett.* 536 (2003) 85–91, [https://doi.org/10.1016/S0014-5793\(03\)00029-2](https://doi.org/10.1016/S0014-5793(03)00029-2).
- [51] D. Hausenloy, D. Yellon, Myocardial ischemia-reperfusion injury: a neglected therapeutic target, *J. Clin. Invest.* 123 (2013) 92–100, <https://doi.org/10.1172/JCI62874.92>.
- [52] E.T. Chouchani, V.R. Pell, E. Gaude, D. Aksentijevic, S.Y. Sundier, E.L. Robb, A. Logan, S.M. Nadtochiy, E.N.J. Ord, A.C. Smith, F. Eayasu, R. Shirley, C. Hu, A. J. Dare, A.M. James, S. Rogatti, R.C. Hartley, S. Eaton, A.S.H. Costa, P.S. Brookes, S.M. Davidson, M.R. Duchon, K. Saeb-parsy, M.J. Shattock, A.J. Robinson, L. M. Work, C. Frezza, T. Krieg, M.P. Murphy, Ischaemic accumulation of succinate controls reperfusion injury through mitochondrial ROS, *Nature* (2014), <https://doi.org/10.1038/nature13909>.
- [53] M.R. Ward, A. Abadeh, K.A. Connelly, Concise review: rational use of mesenchymal stem cells in the treatment of ischemic heart disease, *Stem Cells Transl. Med.* 7 (2018) 543–550, <https://doi.org/10.1002/sctm.17-0210>.

- [54] R. Widyaniangrum, T. Burnouf, O. Nebie, L. Delila, T.J. Wang, A purified human platelet pellet lysate rich in neurotrophic factors and antioxidants repairs and protects corneal endothelial cells from oxidative stress, *Biomed. Pharmacother.* 142 (2021) 112046, <https://doi.org/10.1016/j.biopha.2021.112046>.
- [55] F.G. Kugeratski, K. Hodge, S. Lilla, K.M. McAndrews, X. Zhou, R.F. Hwang, S. Zanivan, R. Kalluri, Quantitative proteomics identifies the core proteome of exosomes with syntenin-1 as the highest abundant protein and a putative universal biomarker, *Nat. Cell Biol.* 23 (2021) 631–641, <https://doi.org/10.1038/s41556-021-00693-y>.
- [56] J.G. White, G. Escolar, The blood platelet open canalicular system: a two-way street, *Eur. J. Cell Biol.* 56 (1991) 233–242.
- [57] R. Suades, T. Padró, G. Vilahur, L. Badimon, Platelet-released extracellular vesicles: the effects of thrombin activation, *Cell. Mol. Life Sci.* 79 (2022), <https://doi.org/10.1007/s00018-022-04222-4>.
- [58] F.G. Gomes, A.C. Andrade, M. Wolf, S. Hochmann, L. Krisch, N. Maeding, C. Regl, R. Poupardin, P. Ebner-Peking, C.G. Huber, N. Meisner-Kober, K. Schallmoser, D. Strunk, Synergy of human platelet-derived extracellular vesicles with secretome proteins promotes regenerative functions, *Biomedicines* 10 (2022) 238, <https://doi.org/10.3390/biomedicines10020238>.
- [59] A. Eulalio, M. Mano, M.D. Ferro, L. Zentilin, G. Sinagra, S. Zacchigna, M. Giacca, Functional screening identifies miRNAs inducing cardiac regeneration, *Nature* 492 (2012) 376–381, <https://doi.org/10.1038/nature11739>.
- [60] F. Gao, M. Kataoka, N. Liu, T. Liang, Z.P. Huang, F. Gu, J. Ding, J. Liu, F. Zhang, Q. Ma, Y. Wang, M. Zhang, X. Hu, J. Kyselovic, X. Hu, W.T. Pu, J. Wang, J. Chen, D. Z. Wang, Therapeutic role of miR-19a/19b in cardiac regeneration and protection from myocardial infarction, *Nat. Commun.* 10 (2019), <https://doi.org/10.1038/s41467-019-09530-1>.
- [61] H.K. Awada, M.P. Hwang, Y. Wang, Towards comprehensive cardiac repair and regeneration after myocardial infarction: aspects to consider and proteins to deliver, *Biomaterials* 125 (2016) 94–112, <https://doi.org/10.1016/j.biomaterials.2015.12.025>.
- [62] Y. Tomioka, T. Kawamura, E. Ito, M. Takeda, A. Harada, K.T. PhD, R. Sakanishi, Y. Sawa, S. Miyagawa, Pleiotropic effects of extracellular vesicles from induced pluripotent stem cell-derived cardiomyocytes on ischemic cardiomyopathy: a pre-clinical study, *J. Heart Lung Transplant.* (2023) 1–15, <https://doi.org/10.1016/j.healun.2023.08.011>.
- [63] E.K.C. Symonds, B. Black, A. Brown, I. Meredith, M.J. Currie, K.E. Hally, K. M. Danielson, Adipose derived stem cell extracellular vesicles modulate primary human macrophages to an anti-inflammatory phenotype in vitro, *J. Extracell. Biol.* 2 (2023), <https://doi.org/10.1002/jex2.104>.
- [64] D.J. Lundy, K.-H. Chen, E.K.-W. Toh, P.C.-H. Hsieh, Distribution of systemically administered nanoparticles reveals a size-dependent effect immediately following cardiac ischaemia-reperfusion injury, *Sci. Rep.* 6 (2016) 25613, <https://doi.org/10.1038/srep25613>.
- [65] J. Li, S. Hu, D. Zhu, K. Huang, X. Mei, B.L.J. de Abad, K. Cheng, All roads lead to rome (The heart): cell retention and outcomes from various delivery routes of cell therapy products to the heart, *J. Am. Heart Assoc.* 10 (2021), <https://doi.org/10.1161/JAHA.120.020402>.
- [66] A. Ciullo, C. Li, L. Li, K.C. Ungerleider, K. Peck, E. Marbán, A.G.E. Ibrahim, Biodistribution of unmodified cardiosphere-derived cell extracellular vesicles using single RNA tracing, *J. Extracell. Vesicles* 11 (2022), <https://doi.org/10.1002/jev2.12178>.
- [67] A. Matsumoto, Y. Takahashi, H.Y. Chang, Y.W. Wu, A. Yamamoto, Y. Ishihama, Y. Takakura, Blood concentrations of small extracellular vesicles are determined by a balance between abundant secretion and rapid clearance, *J. Extracell. Vesicles* 9 (2020), <https://doi.org/10.1080/20013078.2019.1696517>.
- [68] A. Kopic, K. Benamara, M. Schuster, P. Leidenmüller, A. Bauer, H. Glantschnig, W. Höllriegel, Coagulation phenotype of wild-type mice on different genetic backgrounds, *Lab. Anim* 53 (2019) 43–52, <https://doi.org/10.1177/0023677218811059>.
- [69] J. Kisucka, C.E. Butterfield, D.G. Duda, S.C. Eichenberger, S. Saffaripour, J. Ware, Z.M. Ruggeri, R.K. Jain, J. Folkman, D.D. Wagner, Platelets and Platelet Adhesion Support Angiogenesis while Preventing Excessive Hemorrhage, 2006. www.pnas.org/cgi/doi/10.1073/pnas.0510412103.
- [70] M. Adamiak, G. Cheng, S. Bobis-Wozowicz, L. Zhao, S. Kedracka-Krok, A. Samanta, E. Karnas, Y.T. Xuan, B. Skupien-Rabian, X. Chen, U. Jankowska, M. Girgis, M. Sekula, A. Davani, S. Lasota, R.J. Vincent, M. Sarna, K.L. Newell, O.L. Wang, N. Dudley, Z. Madeja, B. Dawn, E.K. Zuba-Surma, Induced Pluripotent Stem Cell (iPSC)-derived extracellular vesicles are safer and more effective for cardiac repair than iPSCs, *Circ. Res.* 122 (2018) 296–309, <https://doi.org/10.1161/CIRCRESAHA.117.311769>.
- [71] L. Barile, V. Lionetti, E. Cervio, M. Matteucci, M. Gherghiceanu, L.M. Popescu, T. Torre, F. Siclari, T. Moccetti, G. Vassalli, Extracellular vesicles from human cardiac progenitor cells inhibit cardiomyocyte apoptosis and improve cardiac function after myocardial infarction, *Cardiovasc. Res.* 103 (2014) 530–541, <https://doi.org/10.1093/cvr/cvu167>.
- [72] I. Diallo, A. Benmoussa, J. Laugier, A. Osman, W.E. Hitzler, P. Provost, Platelet pathogen reduction Technologies alter the MicroRNA profile of platelet-derived microparticles, *Front. Cardiovasc. Med.* 7 (2020), <https://doi.org/10.3389/fcvm.2020.00031>, 1–9.
- [73] D. Qin, X. Wang, Y. Li, L. Yang, R. Wang, J. Peng, K.E.X. Mu, T. Peng, Q. Han, K. J. Yu, G.C. Fan, MicroRNA-223-5p and -3p Cooperatively suppress necroptosis in ischemic/reperfused hearts, *J. Biol. Chem.* 291 (2016) 20247–20259, <https://doi.org/10.1074/jbc.M116.732735>.
- [74] X.P. Ren, J. Wu, X. Wang, M.A. Sartor, J. Qian, K. Jones, P. Nicolaou, T. J. Pritchard, G.C. Fan, MicroRNA-320 is involved in the regulation of cardiac ischemia/reperfusion injury by targeting heat-shock protein 20, *Circulation* 119 (2009) 2357–2366, <https://doi.org/10.1161/CIRCULATIONAHA.108.814145>.
- [75] L. Qiao, S. Hu, S. Liu, H. Zhang, H. Ma, K. Huang, Z. Li, T. Su, A. Vandergriff, J. Tang, T. Allen, P.-U. Dinh, J. Cores, Q. Yin, Y. Li, K. Cheng, microRNA-21-5p dysregulation in exosomes derived from heart failure patients impairs regenerative potential, *J. Clin. Invest.* 129 (2019) 2237–2250, <https://doi.org/10.1172/JCI123135>.
- [76] H. Zhu, G.C. Fan, Role of microRNAs in the reperfused myocardium towards post-infarct remodelling, *Cardiovasc. Res.* 94 (2012) 284–292, <https://doi.org/10.1093/cvr/cvr291>.
- [77] M.L. Lindsey, K.R. Brunt, J.A. Kirk, P. Kleinbongard, J.W. Calvert, L.E. de Castro Brás, K.Y. DeLeon-Pennell, D.P. Del Re, N.G. Frangogiannis, S. Frantz, R.J. Gumina, G.V. Halade, S.P. Jones, R.H. Ritchie, F.G. Spinale, E.B. Thorp, C.M. Ripplinger, Z. Kassiri, Guidelines for in vivo mouse models of myocardial infarction, *Am. J. Physiol. Heart Circ. Physiol.* 321 (2021), <https://doi.org/10.1152/AJPHEART.00459.2021>. H1056–H1073.
- [78] L. Lin, D.N. Cook, G.P. Wiesehahn, R. Alfonso, B. Behrman, G.D. Cimino, L. Corten, P.B. Damonte, R. Dikeman, K. Dupuis, Y.M. Fang, C. V. Hanson, J.E. Hearst, C. Y. Lin, H.F. Londe, K. Metchette, A.T. Nerio, J.T. Pu, A.A. Reames, M. Rheinschmidt, J. Tessman, S.T. Isaacs, S. Wollowitz, L. Corash, Photochemical inactivation of viruses and bacteria in platelet concentrates by use of a novel psoralen and long-wavelength ultraviolet light, *Transfusion (Paris)* 37 (1997) 423–435, <https://doi.org/10.1046/j.1537-2995.1997.37497265344.x>.
- [79] T. Driedonks, L. Jiang, B. Carlson, Z. Han, G. Liu, S.E. Queen, E.N. Shirk, O. Gololobova, Z. Liao, L.H. Nyberg, G. Lima, L. Paniushkina, M. Garcia-Contreras, K. Schonvisky, N. Castell, M. Stover, S. Guerrero-Martin, R. Richardson, B. Smith, V. Machairaki, C.P. Lai, J.M. Izzi, E.K. Hutchinson, K.A.M. Pate, K.W. Witwer, Pharmacokinetics and biodistribution of extracellular vesicles administered intravenously and intranasally to Macaca nemestrina, *J. Extracell. Biol.* 1 (2022) 1–34, <https://doi.org/10.1002/jex2.59>.
- [80] M. Kang, V. Jordan, C. Blenkiron, L.W. Chamley, Biodistribution of extracellular vesicles following administration into animals: a systematic review, *J. Extracell. Vesicles* 10 (2021), <https://doi.org/10.1002/jev2.12085>.
- [81] Q. Li, Y. Song, Q. Wang, J. Chen, J. Gao, H. Tan, S. Li, Y. Wu, H. Yang, H. Huang, Y. Yu, Y. Li, N. Zhang, Z. Huang, Z. Pang, J. Qian, J. Ge, Engineering extracellular vesicles with platelet membranes fusion enhanced targeted therapeutic angiogenesis in a mouse model of myocardial ischemia reperfusion, *Theranostics* 11 (2021) 3916–3931, <https://doi.org/10.7150/thno.52496>.
- [82] T. Wang, T. Zhou, M. Xu, S. Wang, A. Wu, M. Zhang, Y.L. Zhou, J. Shi, Platelet membrane-camouflaged nanoparticles carry microRNA inhibitor against myocardial ischaemia–reperfusion injury, *J. Nanobiotechnol.* 20 (2022), <https://doi.org/10.1186/s12951-022-01639-8>.
- [83] M.J. Hernandez, R. Gaetani, V.M. Pieters, N.W. Ng, A.E. Chang, T.R. Martin, E. van Ingen, E.A. Mol, M. Dzieciatowska, K.C. Hansen, J.P.G. Sluijter, K.L. Christman, Decellularized extracellular matrix hydrogels as a delivery platform for MicroRNA and extracellular vesicle therapeutics, *Adv. Ther.* 1 (2018) 1800032, <https://doi.org/10.1002/adtp.201800032>.
- [84] U. Tariq, M. Gupta, S. Pathak, R. Patil, A. Dohare, S.K. Misra, Role of biomaterials in cardiac repair and regeneration: therapeutic intervention for myocardial infarction, *ACS Biomater. Sci. Eng.* 8 (2022) 3271–3298, <https://doi.org/10.1021/acsbomaterials.2c00454>.